POLITECNICO DI MILANO

Biochip Course

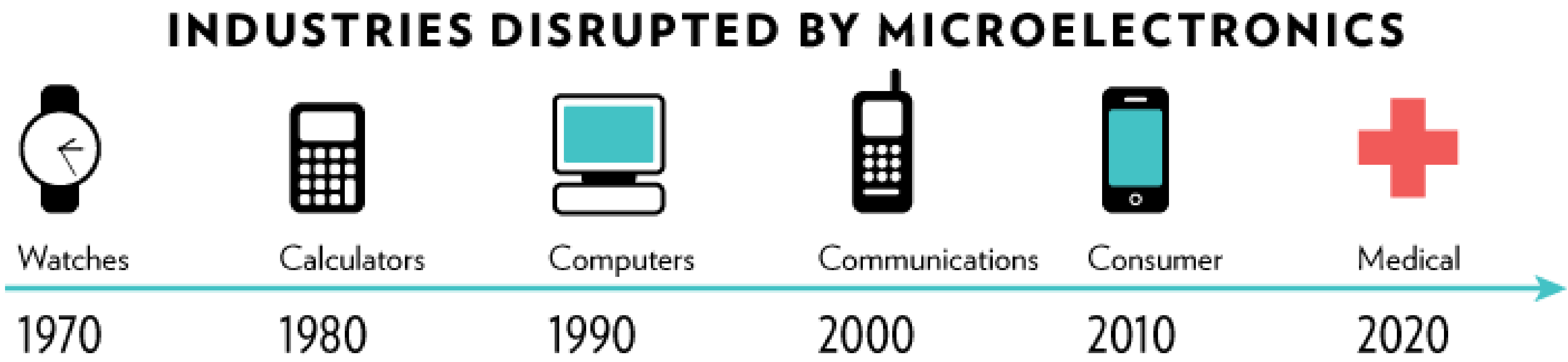
CMOS Biochips

Marco Carminati

a.y. 2022/2023

May 8th 2023

Why Integrated Electronics?



1) Extremely compact (embedded in lab-on-chip device)
2) Highly parallel (thousands of sensing channels)
3) Extremely low-cost (*only if mass production!*)
4) Good performance (close to the sensor)
5) Compatible with microcomputers and wireless

Smart Systems

Critical CMOS Biochip Design Aspects

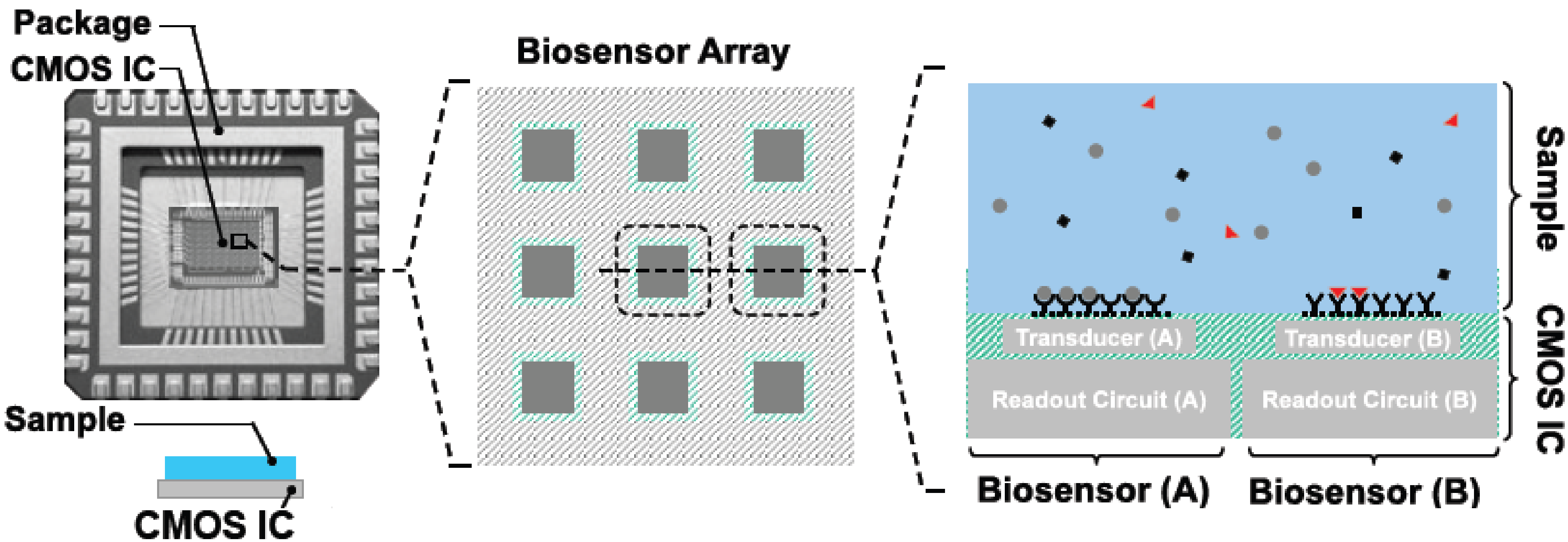
Tighter **constraints** with respect to discrete components:

- Limitations in the circuit components (no large R, C, L)
- Power dissipation - thermal budget

For electrochemical sensors:

- Electrodes: post-CMOS metallization of the topmost metal layer
- Issues due to the conductive silicon substrate (stray impedance)
- Packaging (small size, waterproof, automatable)

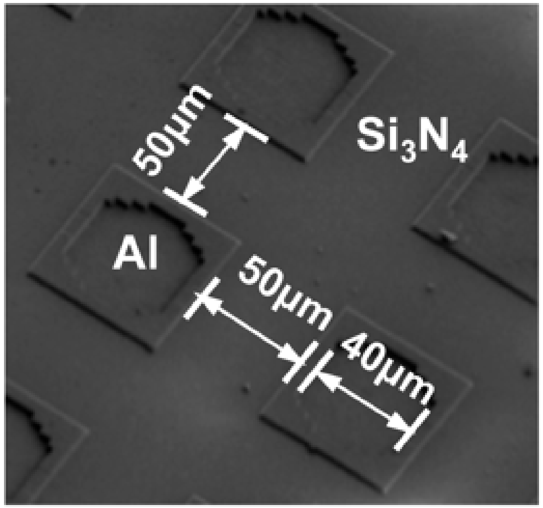
Biosensor Array Architecture

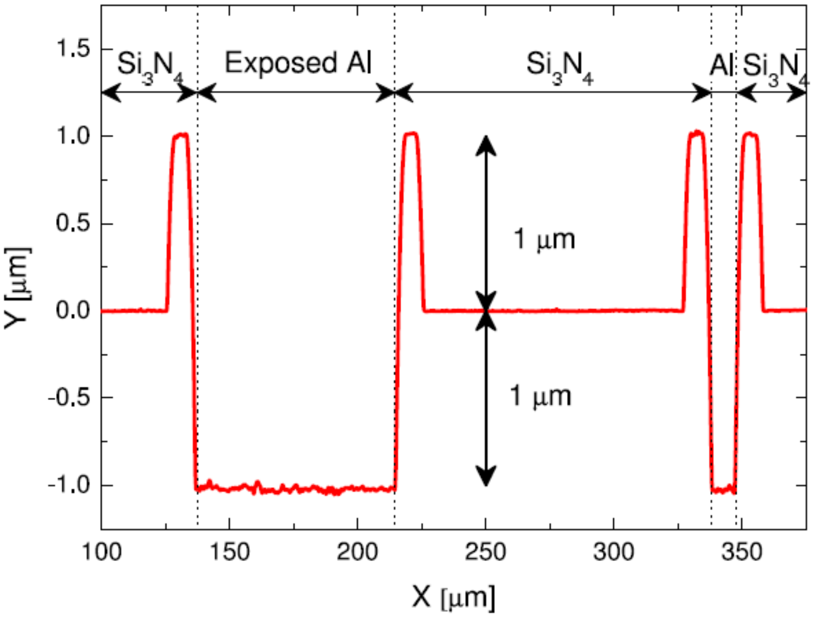


Aluminum of metal lines and bonding pad is too reactive and must be «converted» into gold or platinum for electrochemical detection.

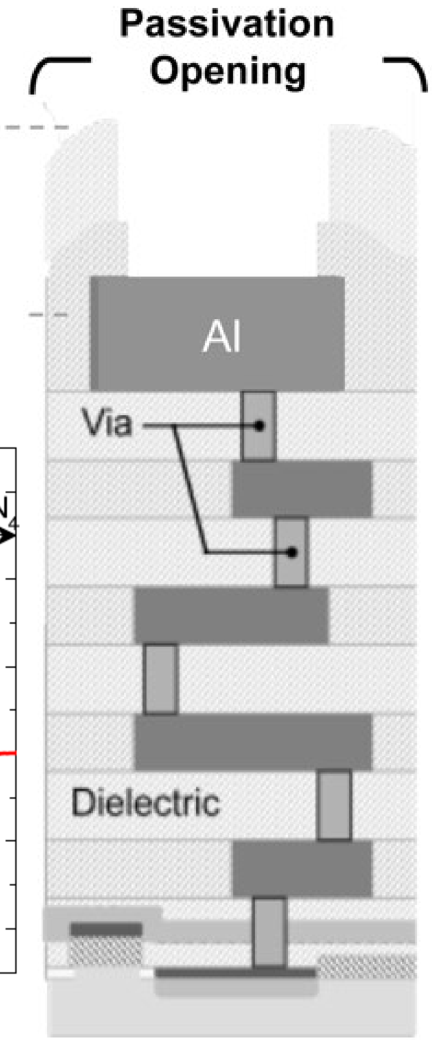
The top passivation (Si_3N_4) is waterproof.

Electrodes: Openings in the Chip Passivation

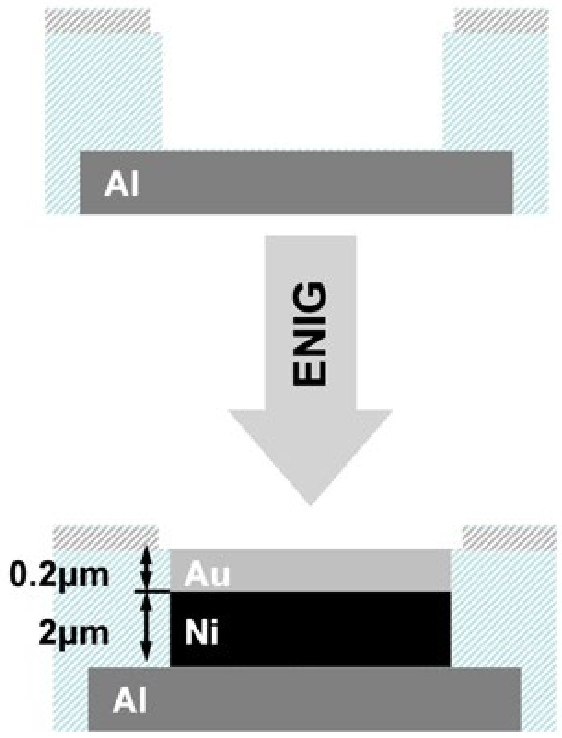




CMOS Cross-section

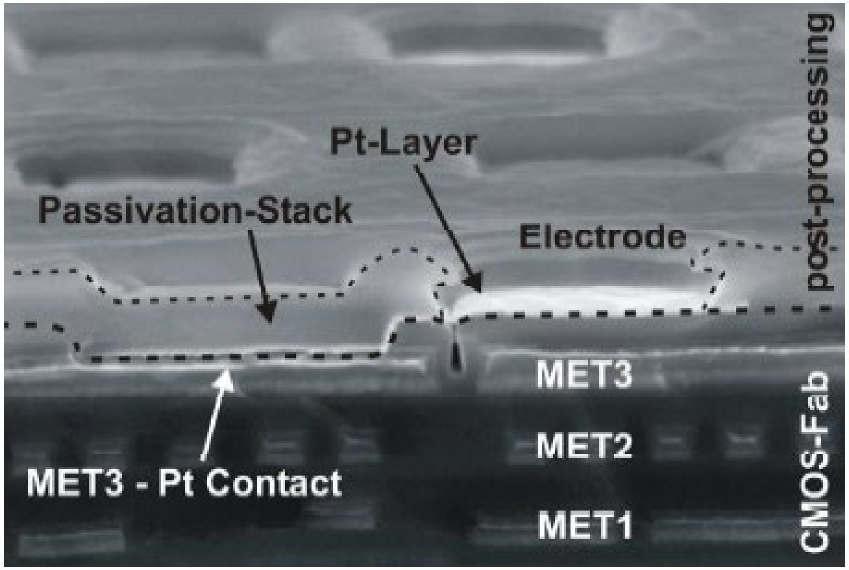


CMOS Electrode

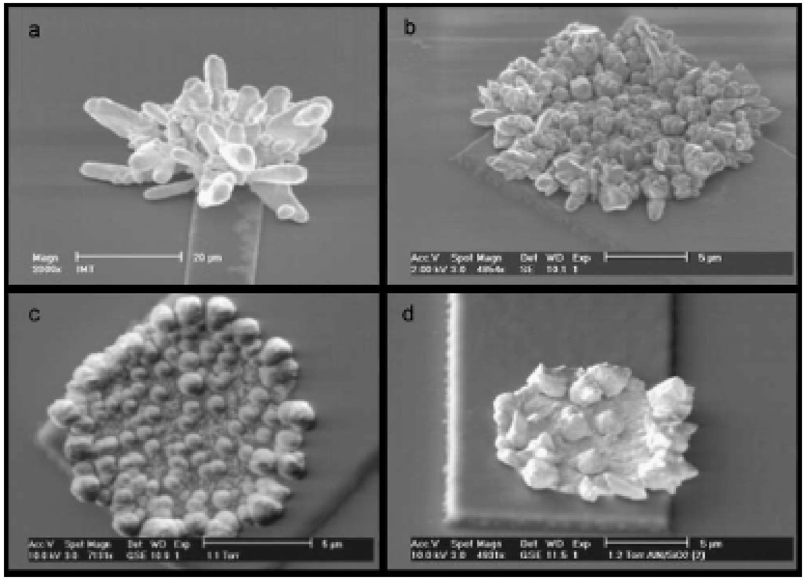


CMOS Post-Processing Electrode Metallization

In the clean-room



On the bench



Accurate (expensive)

Limitation on materials

Temperature limitations:

Chip: 450°C PDMS: 343°C

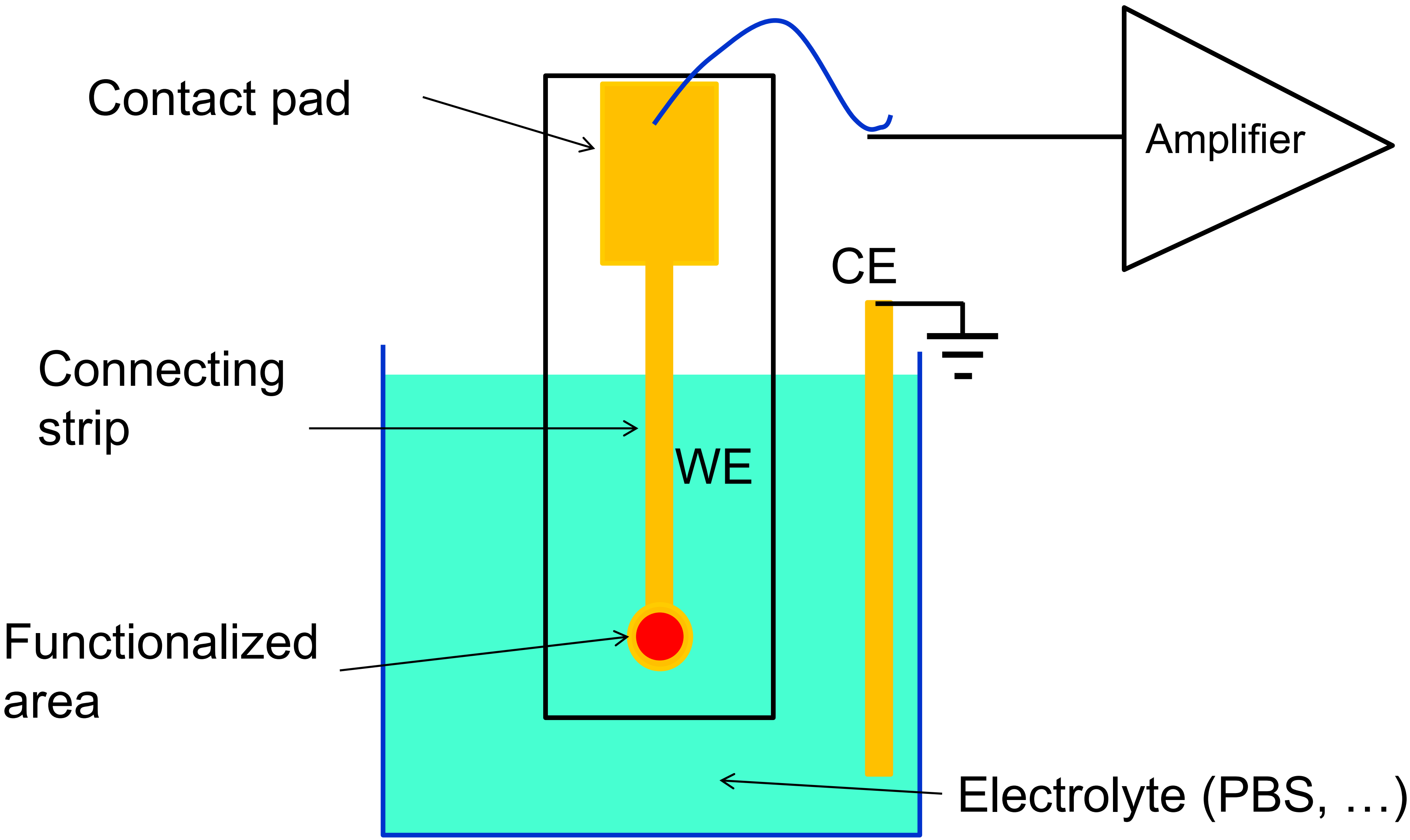
SU8: 200°C Glue: 340°C

Electroplating

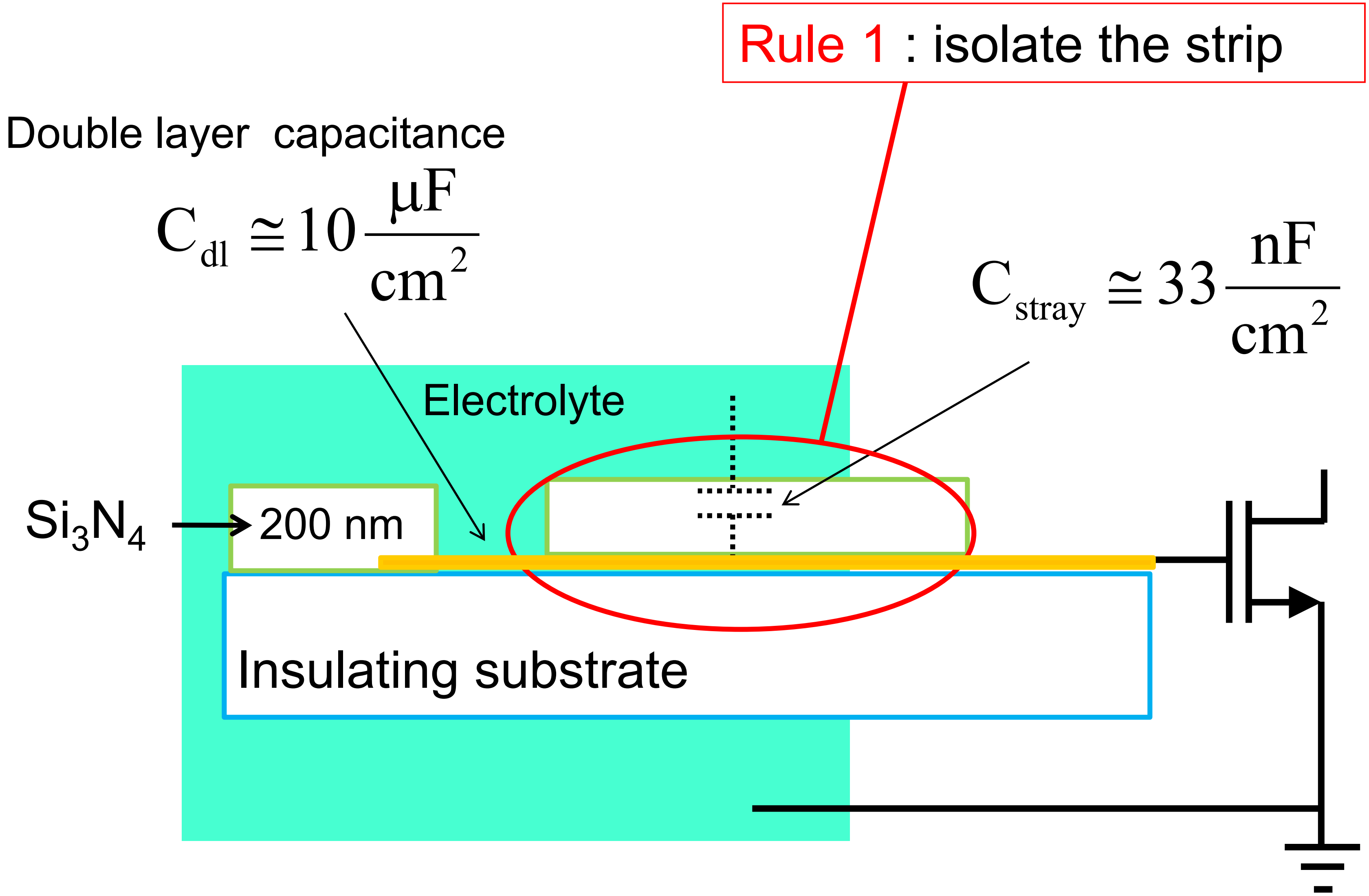
High roughness

The Role of the Substrate

Case Study: Disposable Cartridge



Insulating Substrate

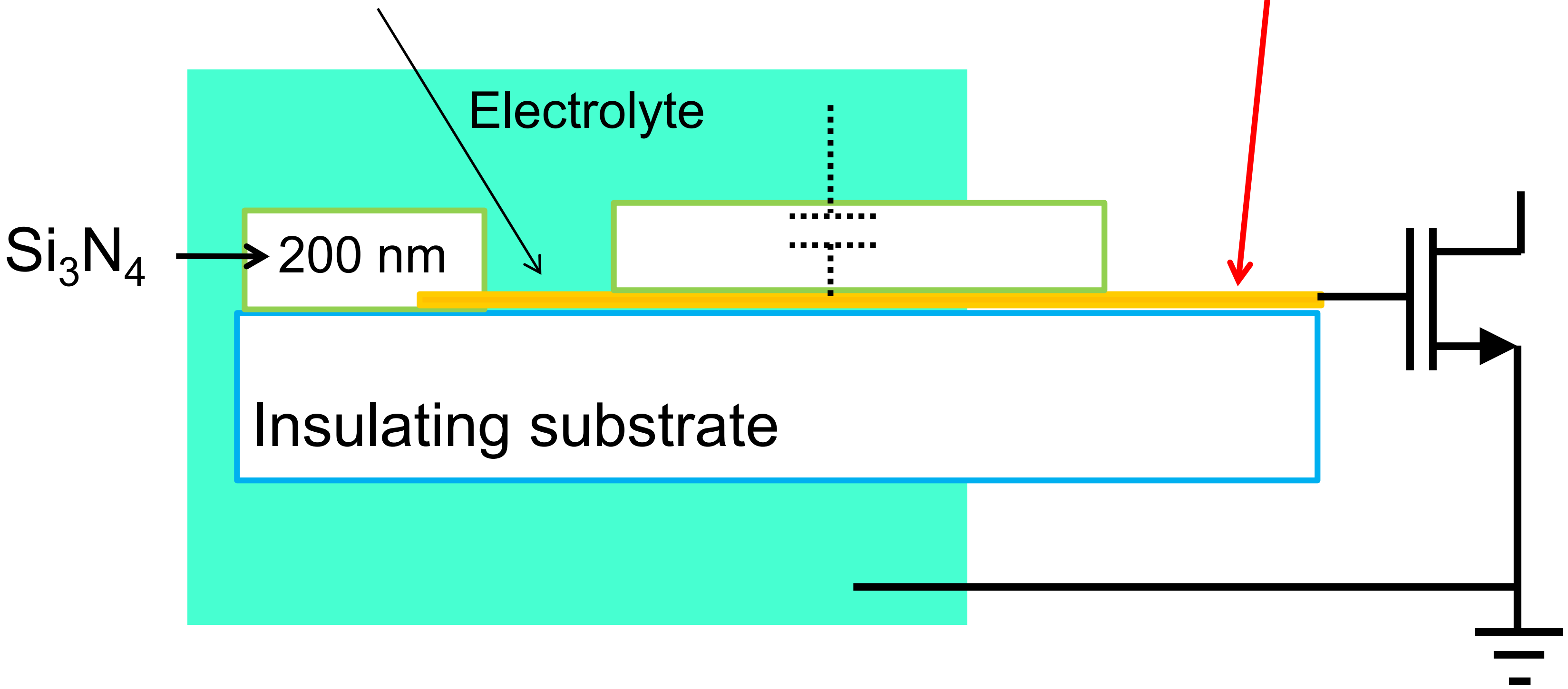


Insulating Substrate

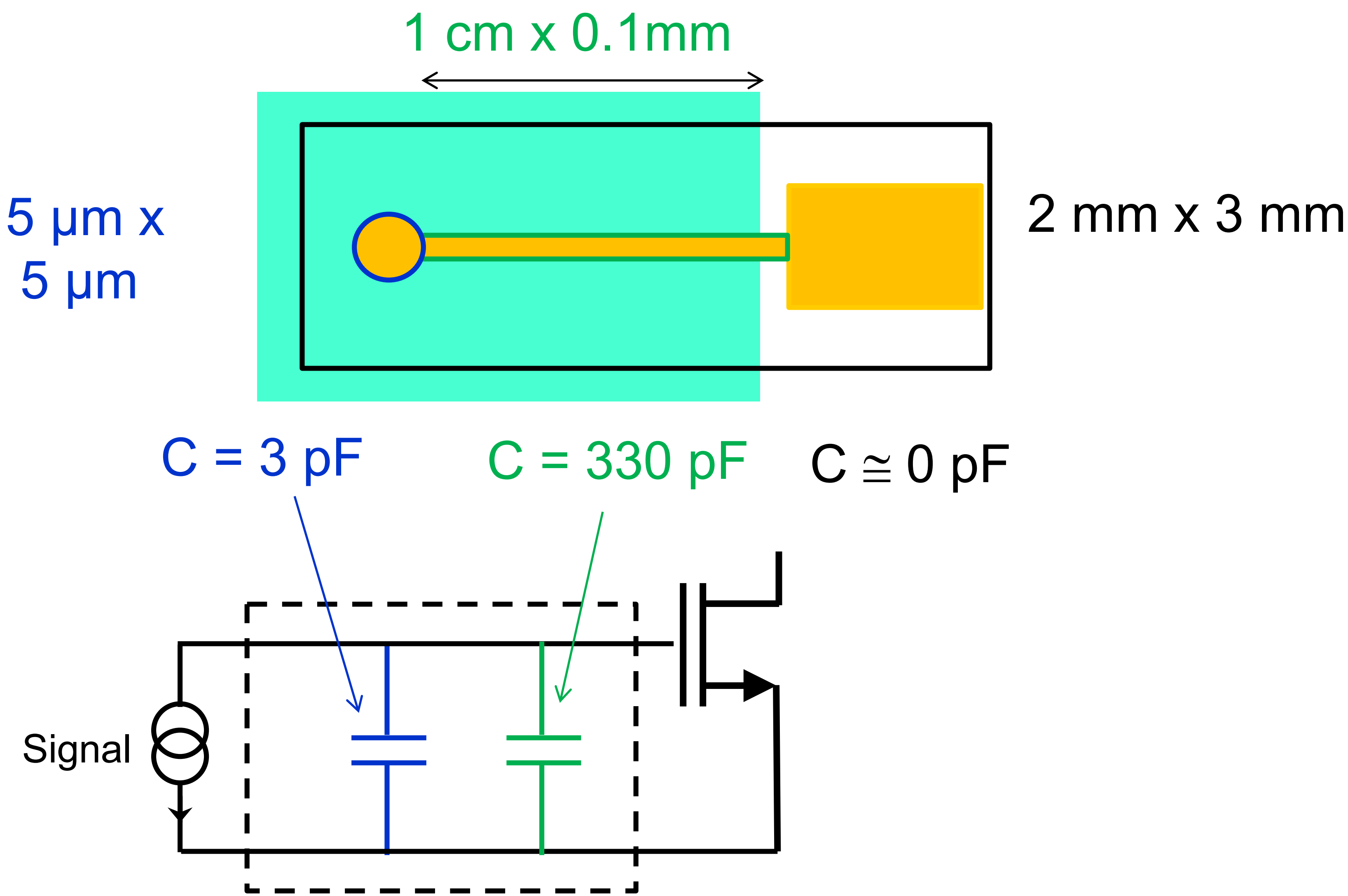
Double layer capacitance

$$C_{dl} \cong 10 \frac{\mu F}{cm^2}$$

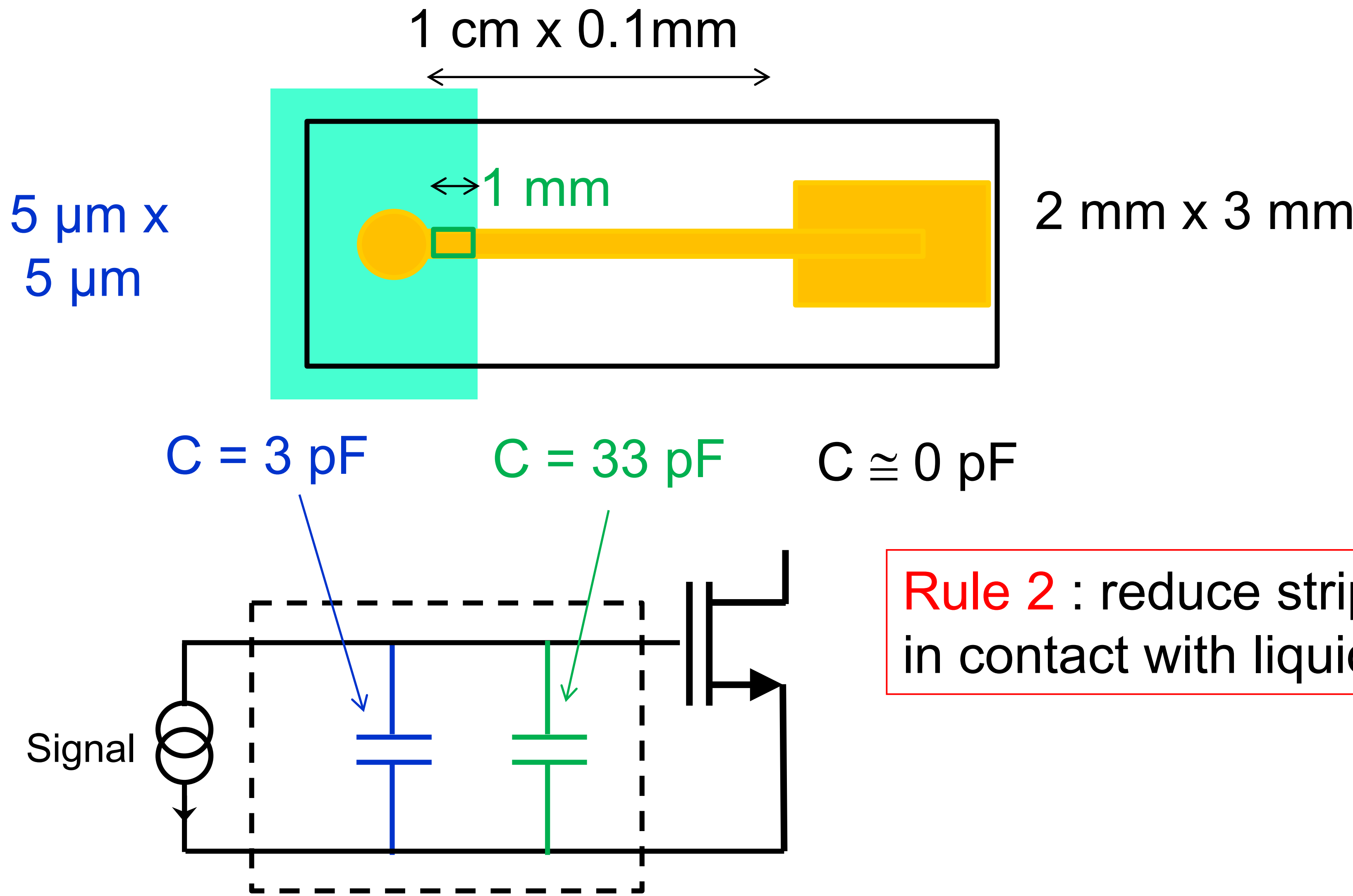
Contact pad
is NOT contributing



Insulating Substrate

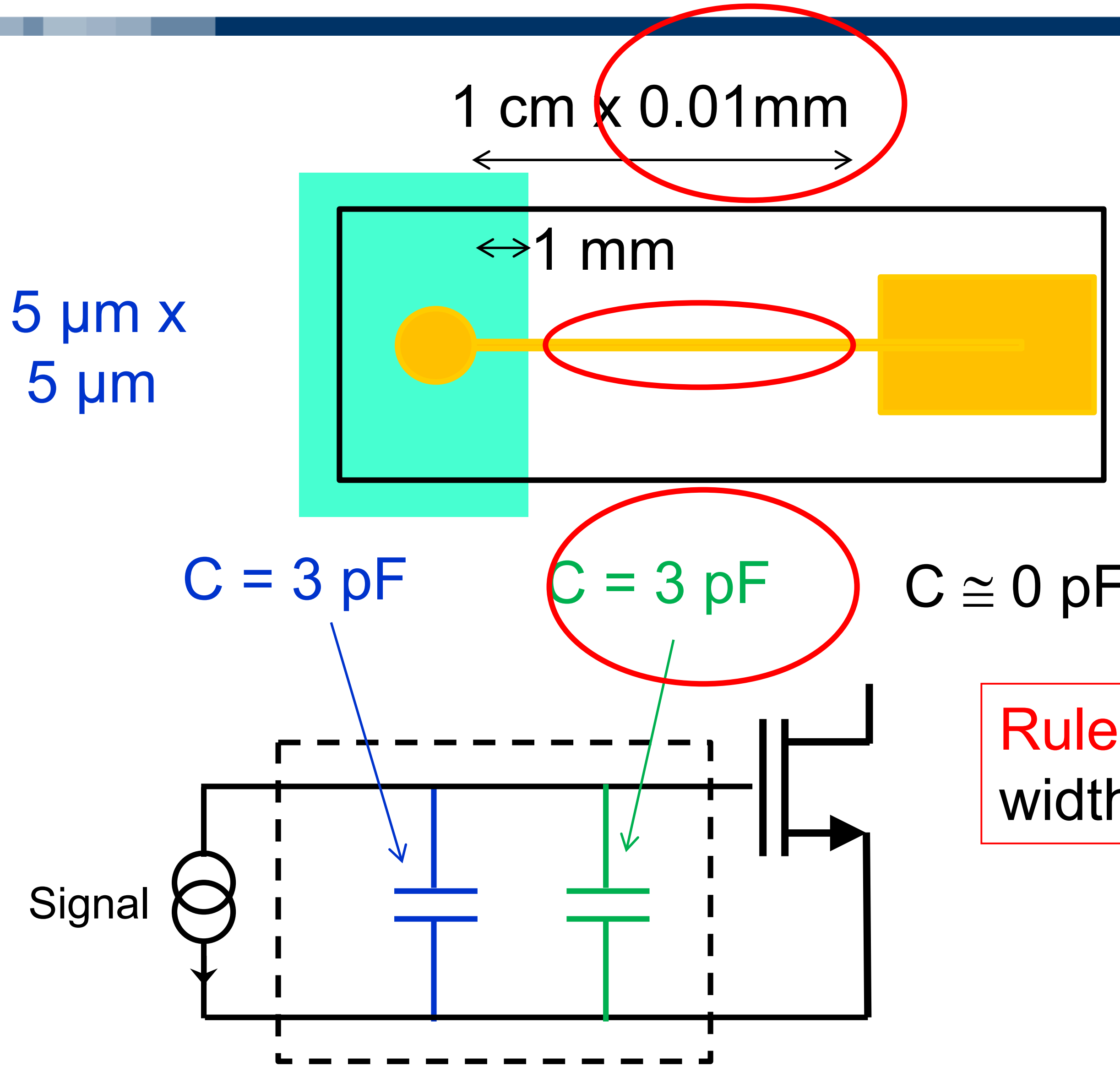


Insulating Substrate



Rule 2 : reduce strip in contact with liquid

Insulating Substrate

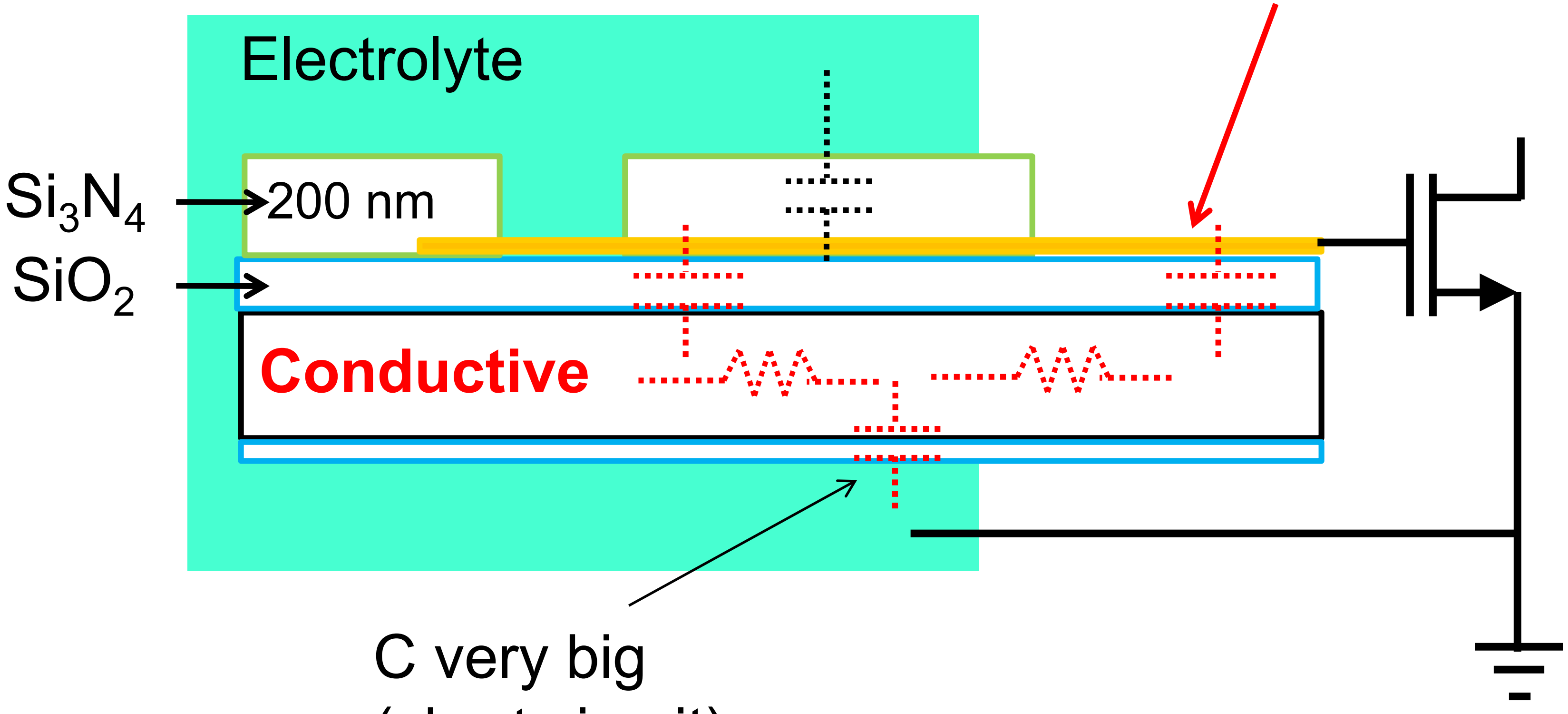


Rule 3 : reduce width of the strip

Conductive Substrate

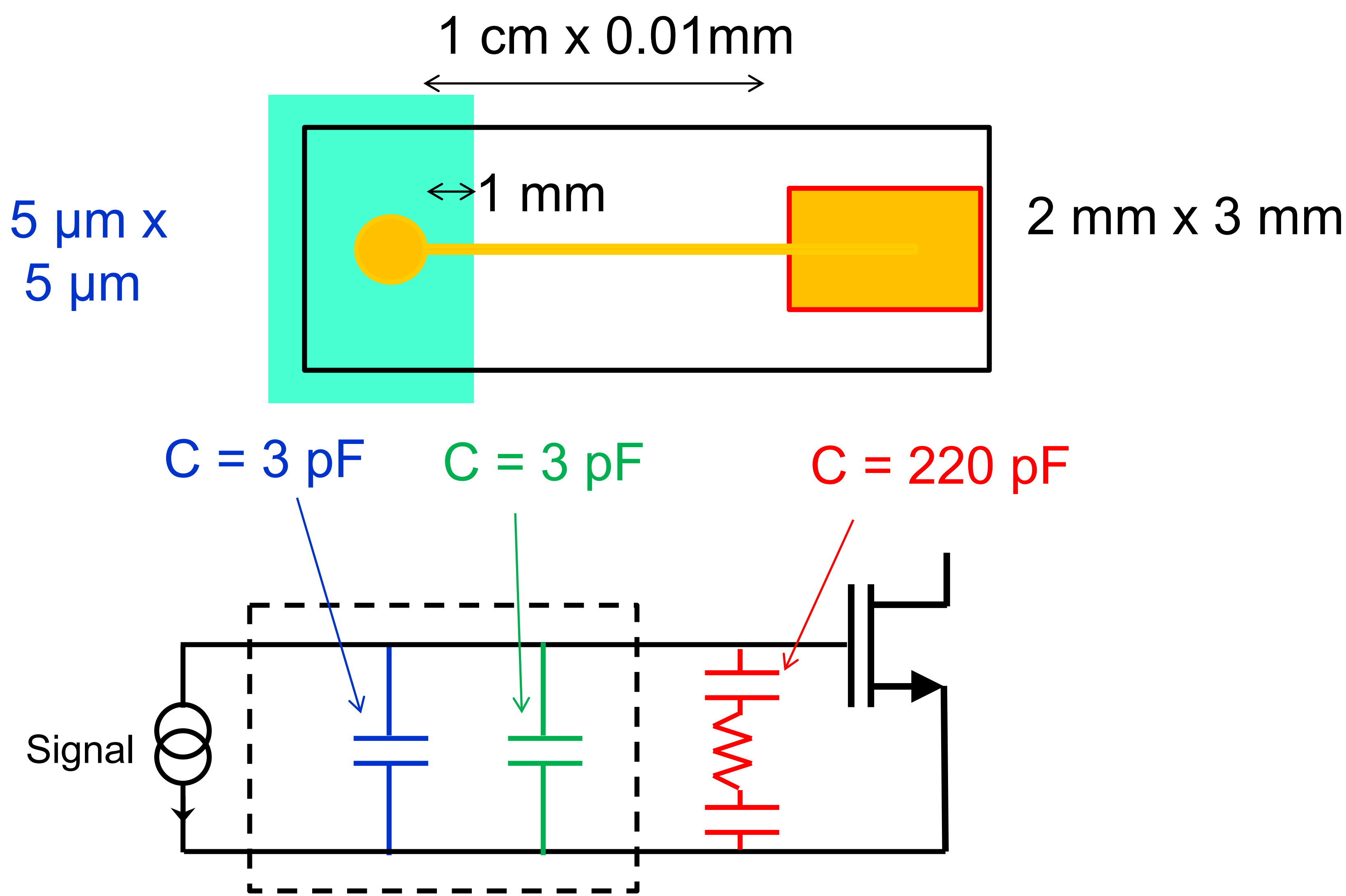
Rule 4 : try NOT to use conductive substrates

Contact pad is contributing !

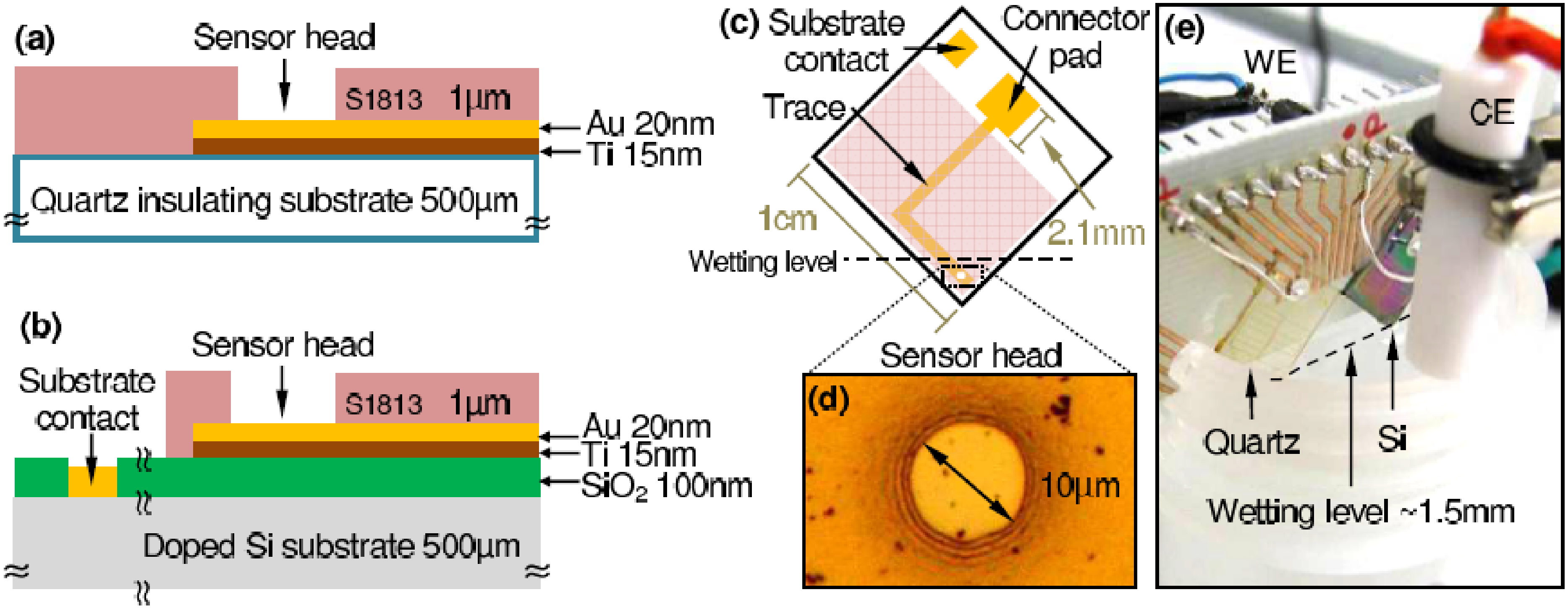


C very big
(short circuit)

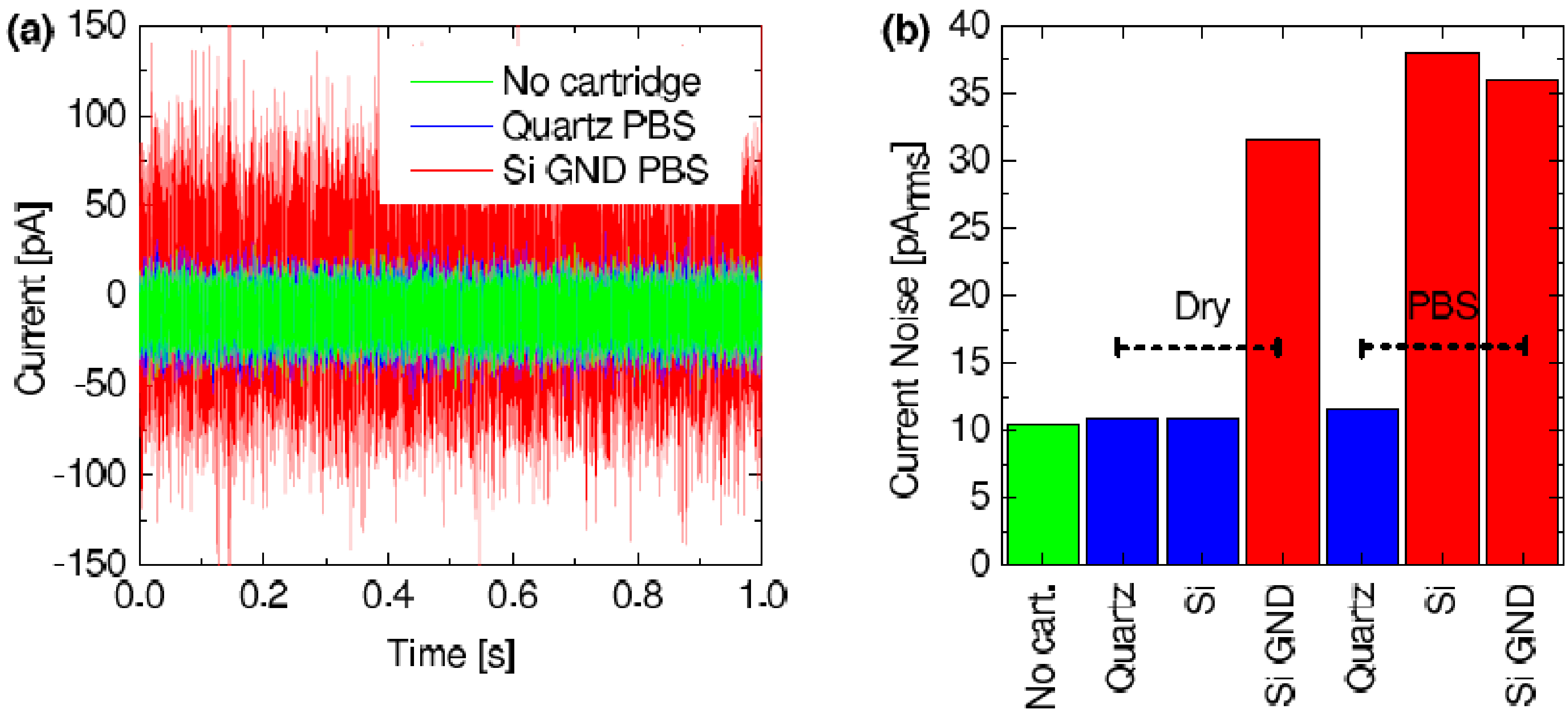
Conductive Substrate



Direct Experimental Comparison



Direct Experimental Comparison



Conclusion: for passive chips Si should be avoided. For active CMOS biochips, the effect of the substrate must be taken into account and mitigated with proper process and geometry choices

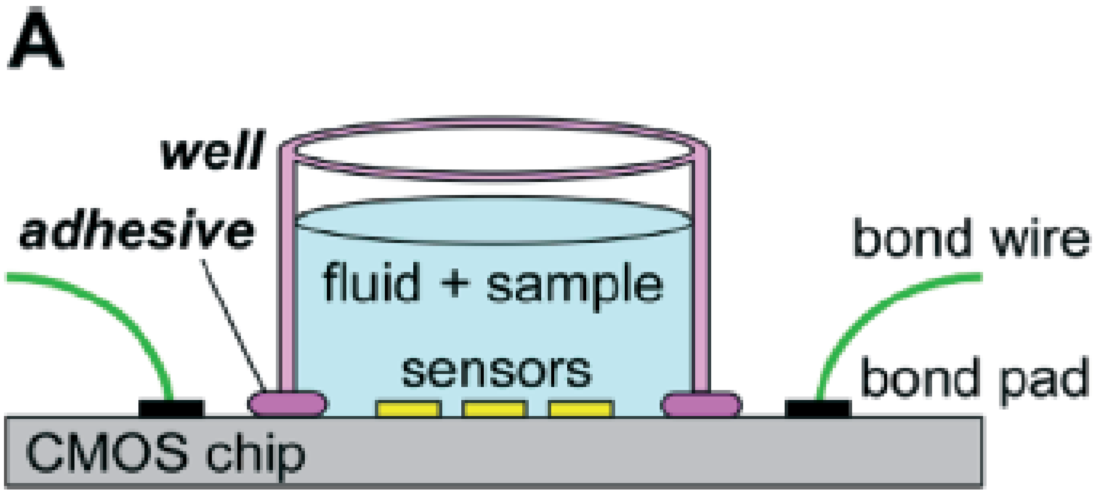
Packaging

A suitable packaging should provide:

- Protection of the chip from the liquid
- Protection of the bonding wires (without breaking them)
- Patternable exposure of the electrodes to the liquid
- Long-term sealing for re-usable chips
- Connectors/interface with the rest of the system

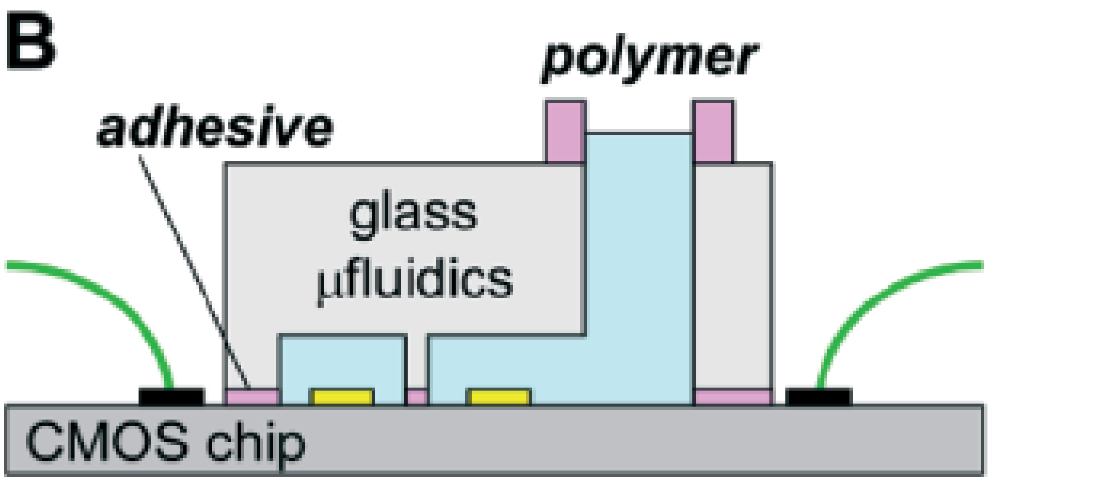
A standard biochip packaging approach is still missing. Several artisanal solutions have been proposed. Industry need an automatable approach.

Packaging Approaches



Simplest

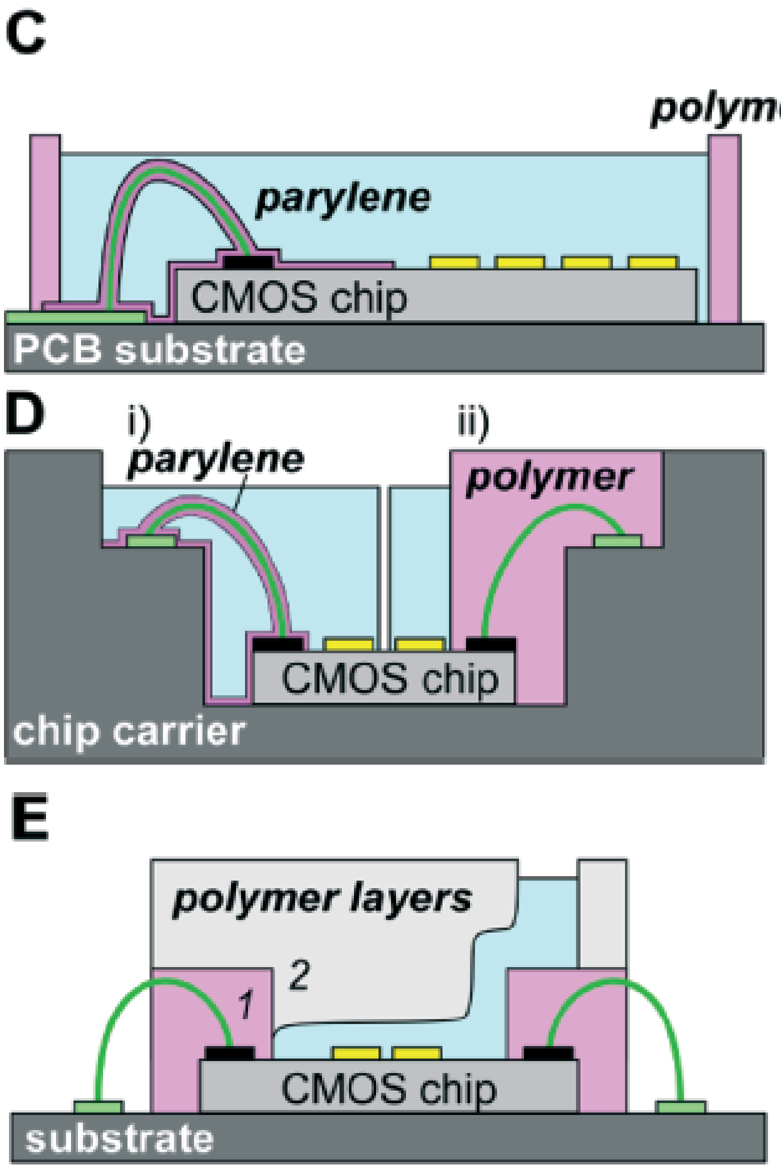
For large chips, but Si area is expensive



Rigid microfluidics

Sealing issues

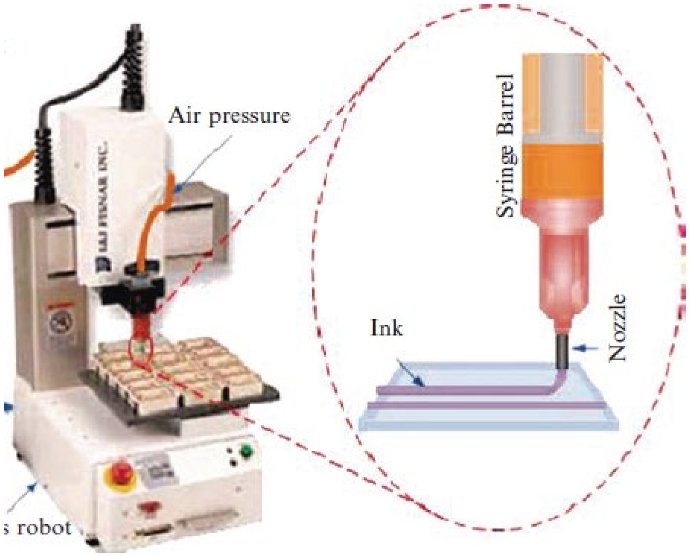
Packaging Approaches

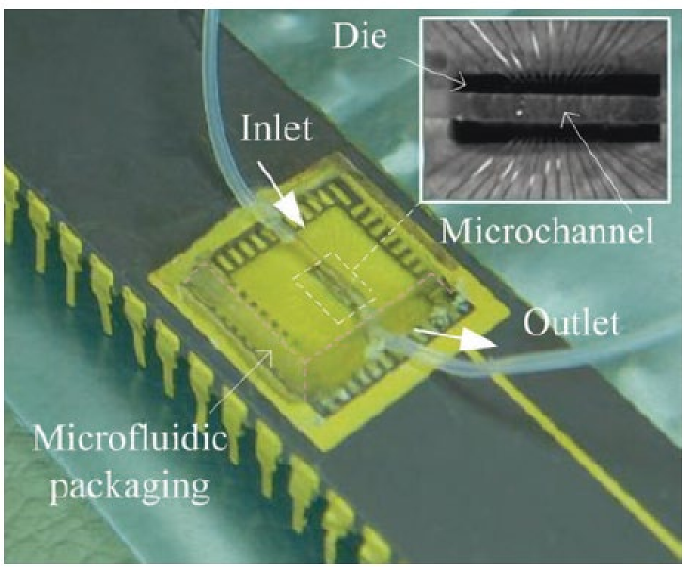


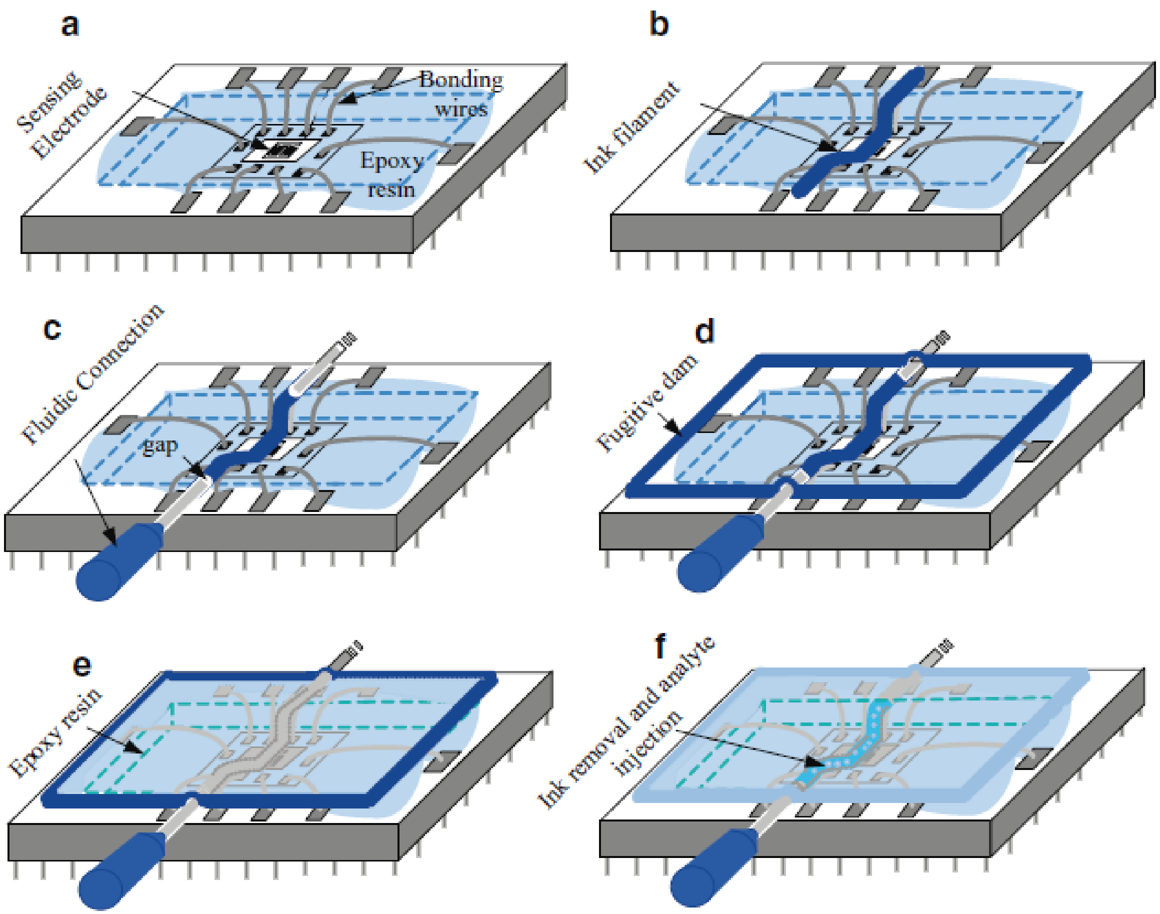
Cover with photo-patternable soft passivation

Sacrificial layer

Direct 3D Printing of Sacrificial Channel







Figures of Merit for Comparison

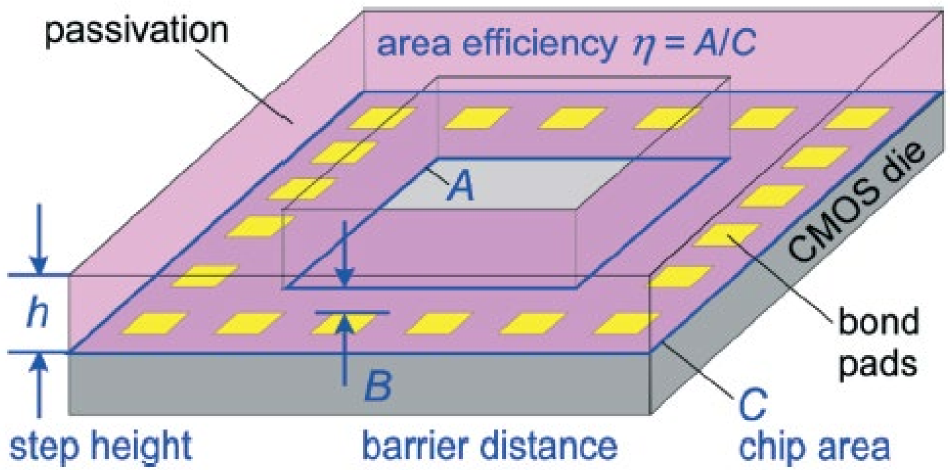


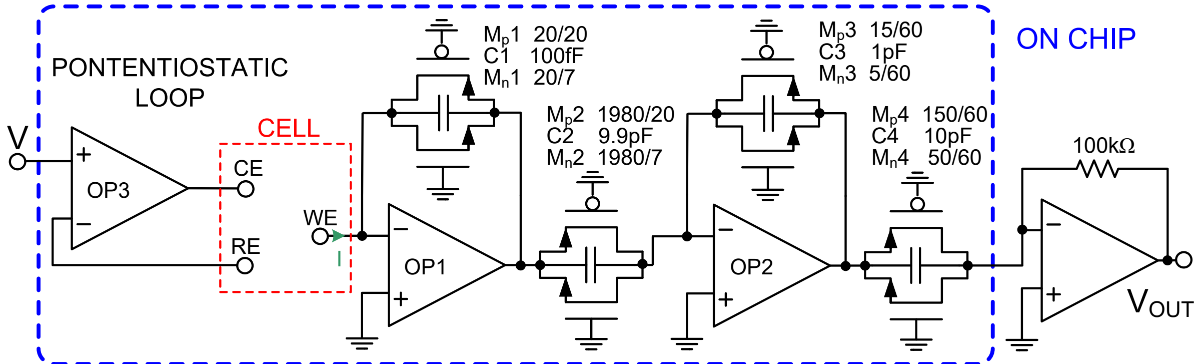
Table 1 Evaluation of the packaging methods that have been used for integrating CMOS chips with microfluidics. Each of the packaging methods in Fig. 7 is represented except for those not done using active chips or that remain theoretical. Entries were inferred or estimated (indicated by *) whenever possible from the available information if they were not given explicitly in the paper. (The ESI explains the methods by which the metrics were determined.) Codes: — = not reported, * = estimated, † = not estimated, BP = before packaging, BW/C = bond wire/chip height, F = flip-chip, H = high, L = low, NA = not applicable, TAB = tape automated bonding, TF = thin film, W = wire-bonded

References	Type (Fig. 7)	Chip size (mm)	Bonding method	Sensor type	Fluid barrier material	Area efficiency*, η	Lifetime (days)	Barrier distance (μm)*	Number of steps	Vertical step height, h (μm)	Post-processing possible?	Microfluidics integrated
11	A	Large[a]	W	Magnetic	PDMS/plastic	L	≥10	≤400	≥4*	H	N	N
2[b]	B	5 × 47	W	Optical	SiO_2	H	—	≥500	≥9	NA	BP	Y
34	C	—	W	ISFET	Parylene	H	H*	<100	4*	300*	N	N
5	D	4.4 × 4.4	W	Electrical	Epoxy & PDMS	L	≥28	>1000	≥6*	H	BP	N
25	D	6.5 × 6.5	W	Electrical	Epoxy & PDMS	L	≥56	>1000	≥6*	H	BP	N
41	D	3 × 3	W	Electrochemical	Parylene	H	H*	—	≥6*	BW/C	N	N
40	D	3.2 × 3.2	W	Electrical	Silastic 9161 RTV	H	≥14	≥200	≥8*	H	BP	Y
40	D	3.2 × 3.2	W	Electrical	Hysol CB064	H	≥14	≥200	>8*	H	BP	Y
50	D	5.2 × 6.5	W	Electrical	Epoxy	L	≥1	≥1000	≥4*	H	BP	N
12	E	2 × 5	W	Magnetic & temperature	SU-8	L	—	>1000	≥6*	Low*	Y	Y
14	E	1 × 2.4	W	Optical	Epoxy & PDMS	H	L*	≥400	≥4*	BW/C	N	Y
45	E	1 × 2*	W	Capacitive	Epoxy	L	—	NA	≥5*	BW/C	N	Y
6	E	1 × 4	W	Magnetic	Polyimide	L	—	≥1000	≥7	BW/C	N	Y
6	E	2 × 5	W	Magnetic	SU-8	L	—	≥1000	≥6	BW/C	N	Y

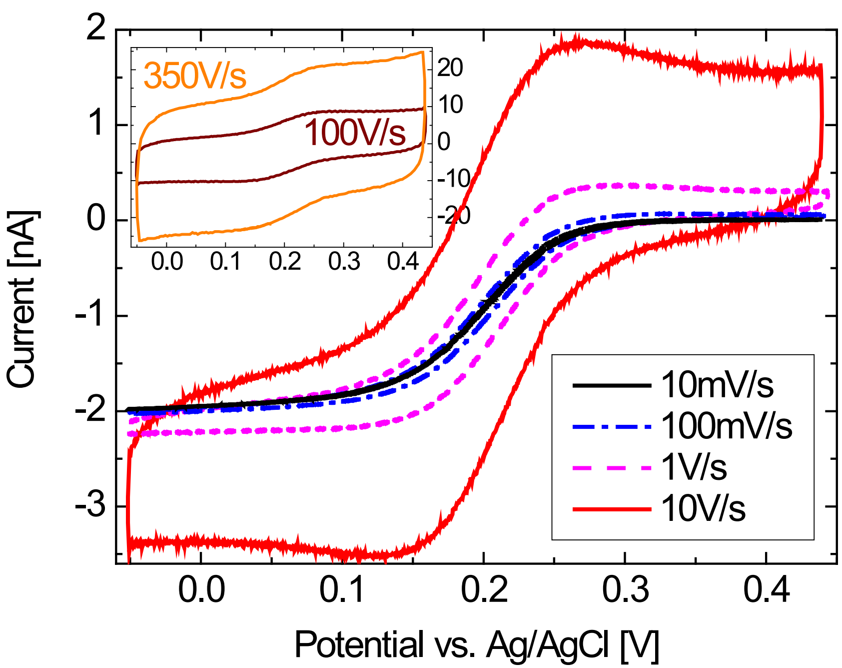
CMOS Biochips State-of-the-Art Examples from the Literature

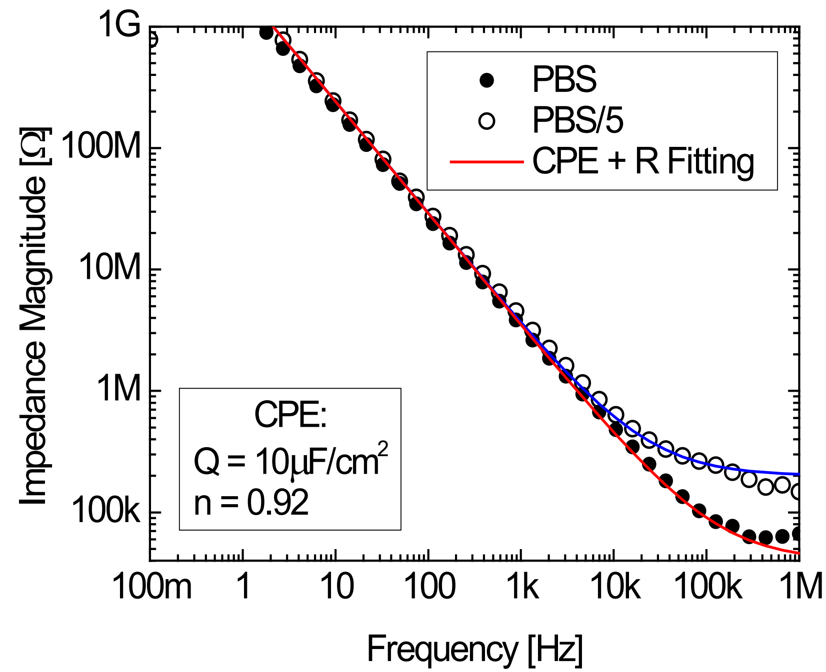
Ex 1: Single-Chip CMOS Potentiostat (1 Ch.)

Single channel, top-level **performance:**

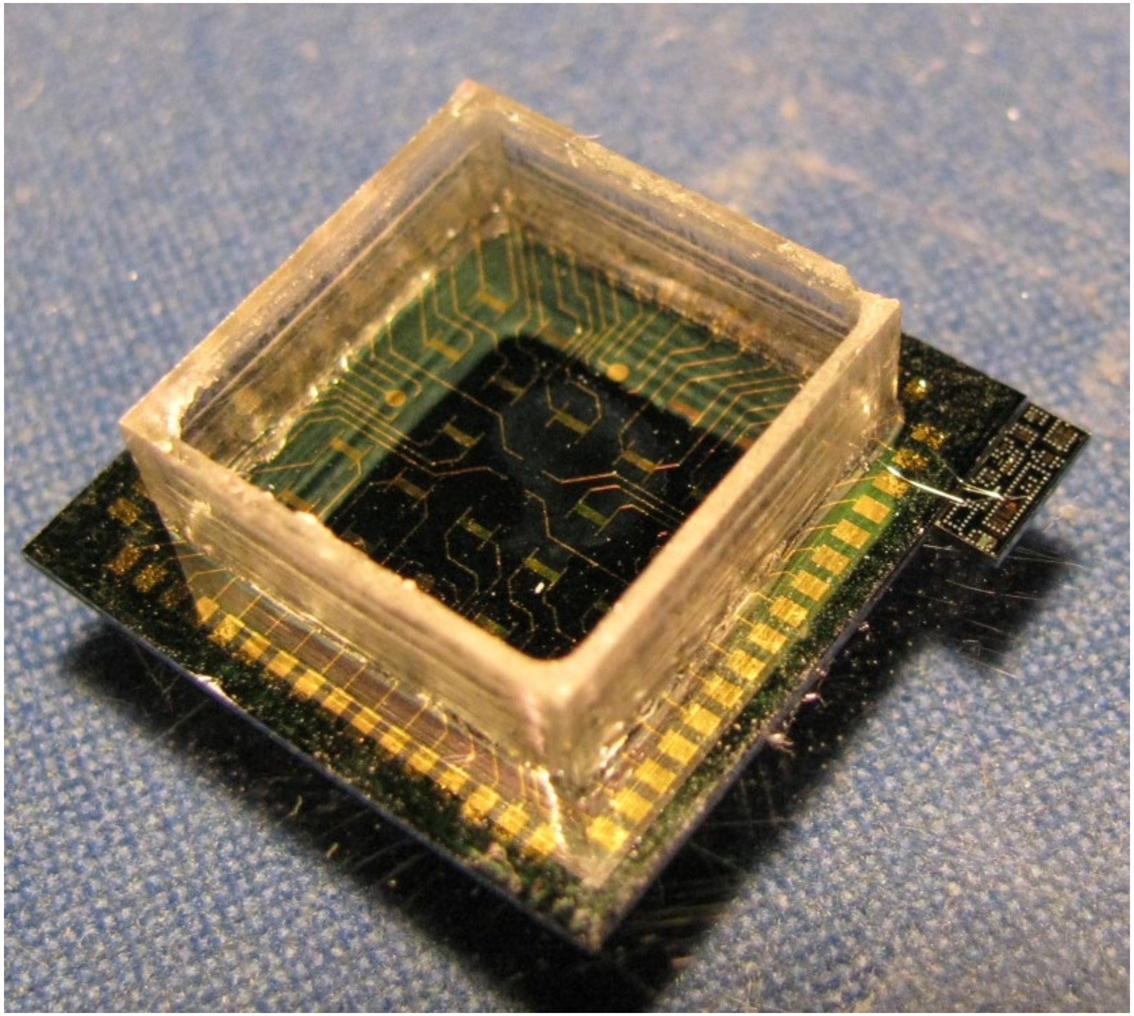


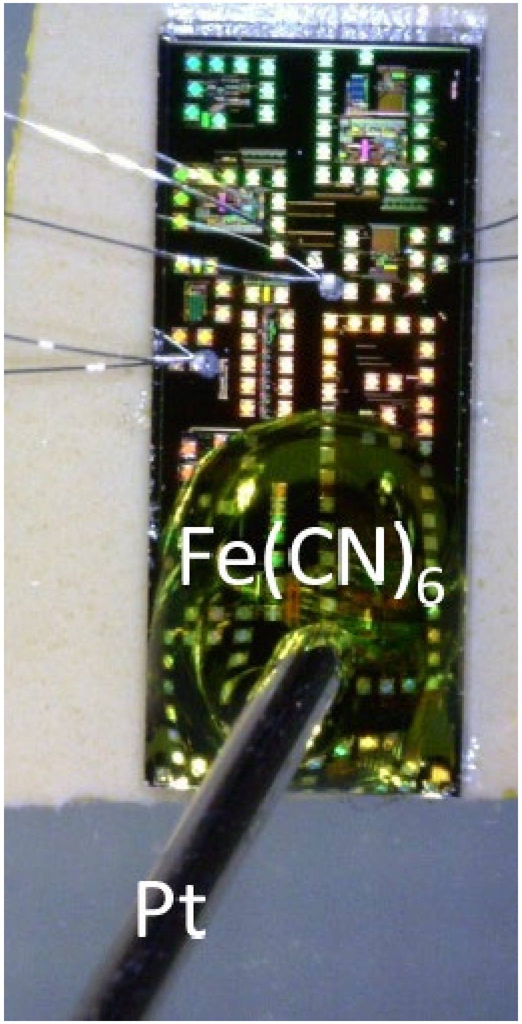
- 0-1MHz BW
- 120dB DR





M. Carminati *et al.*, *IEEE ICECS* 2009

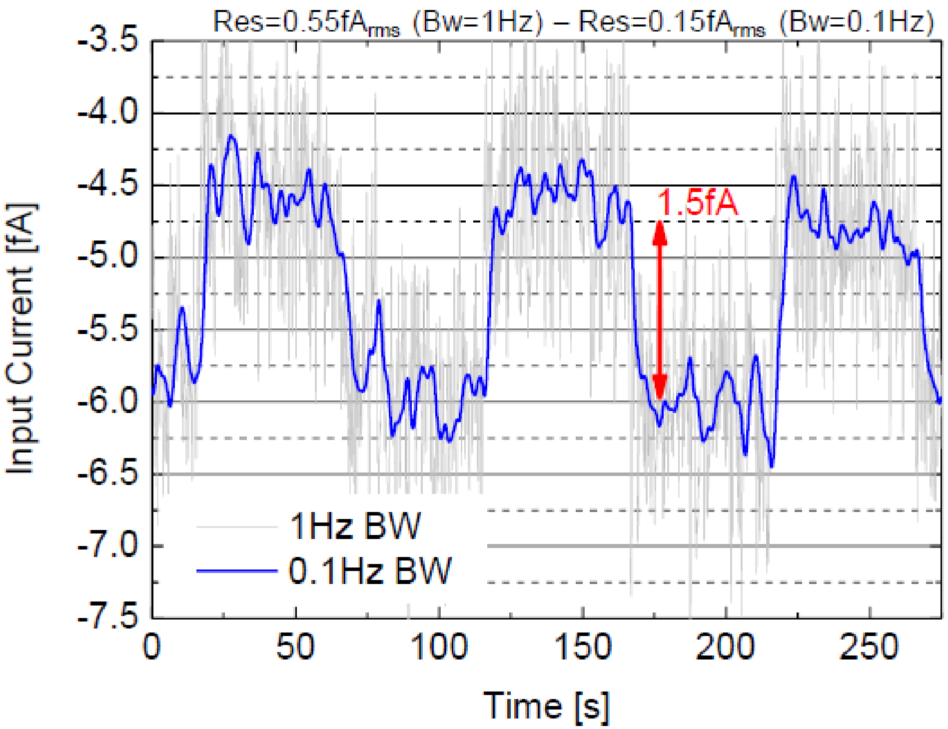



$Fe(CN)_6$
Pt

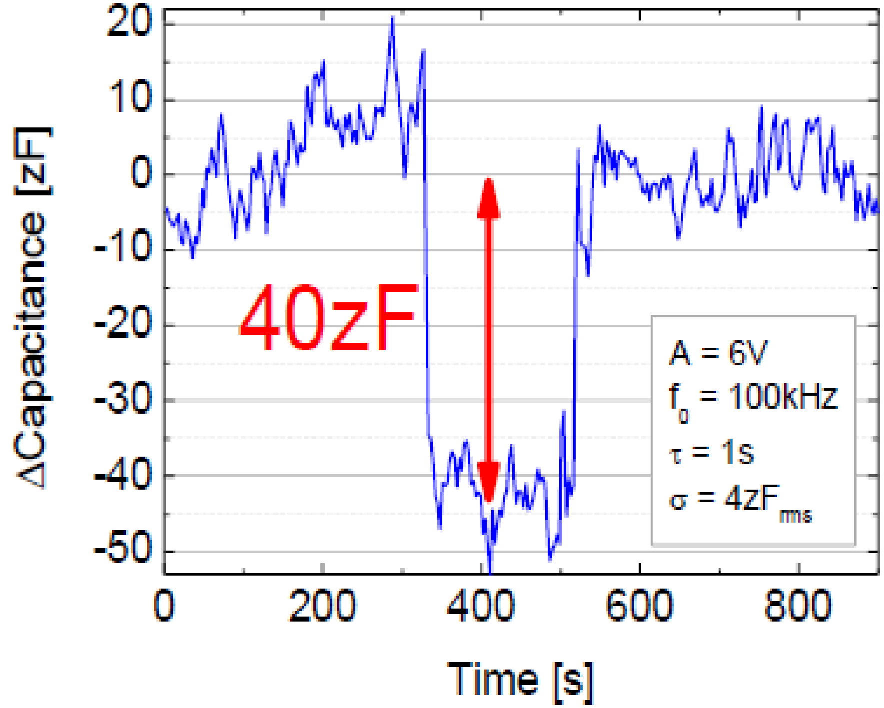
Record Performance

femtoAmpere!

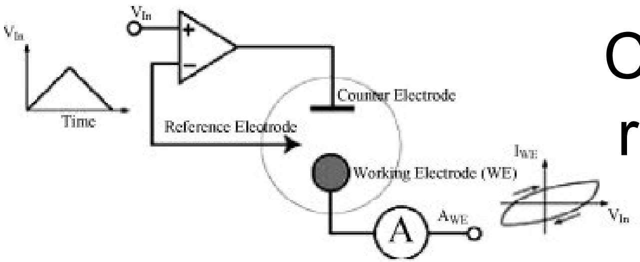
1.5fA current steps (9 electrons in 1ms)



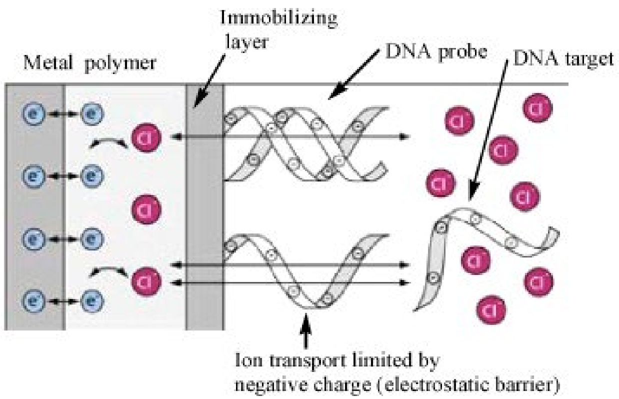
zeptoFarad resolution!!



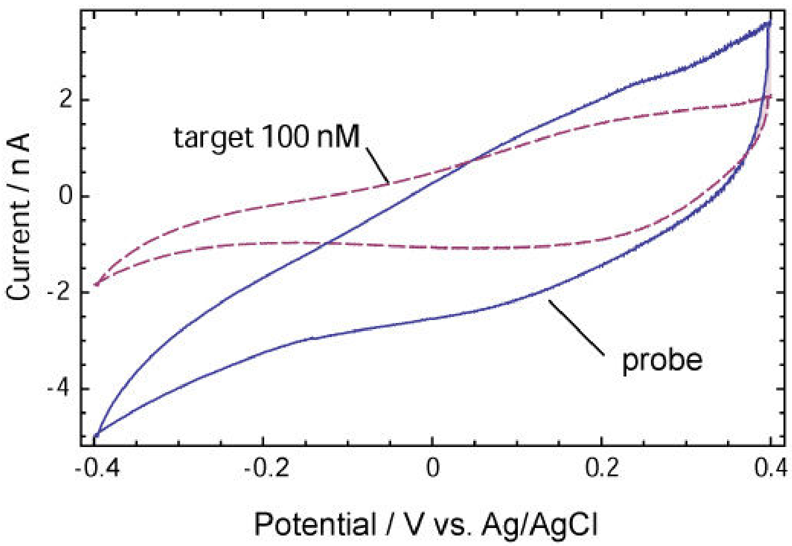
Ex 2: Multichannel Potentiostat (576 Ch.)

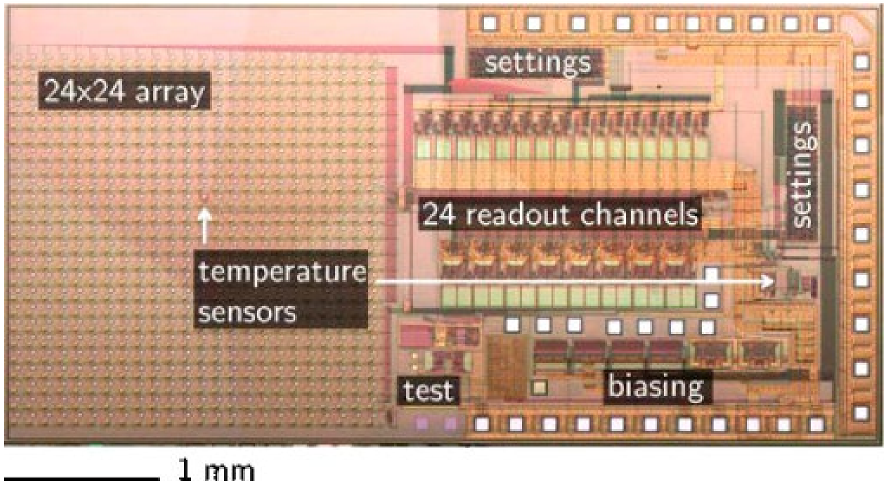


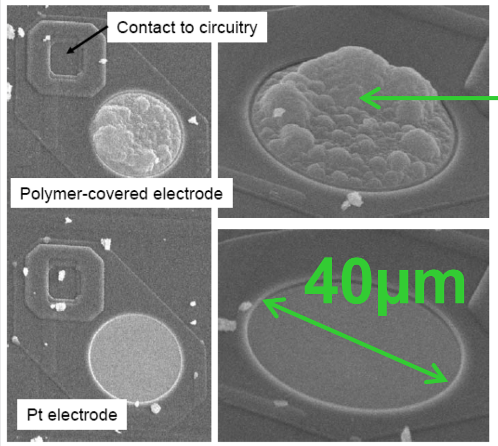
Current reader



100 nM complementary 30-mer
average area change (N=46): -38%



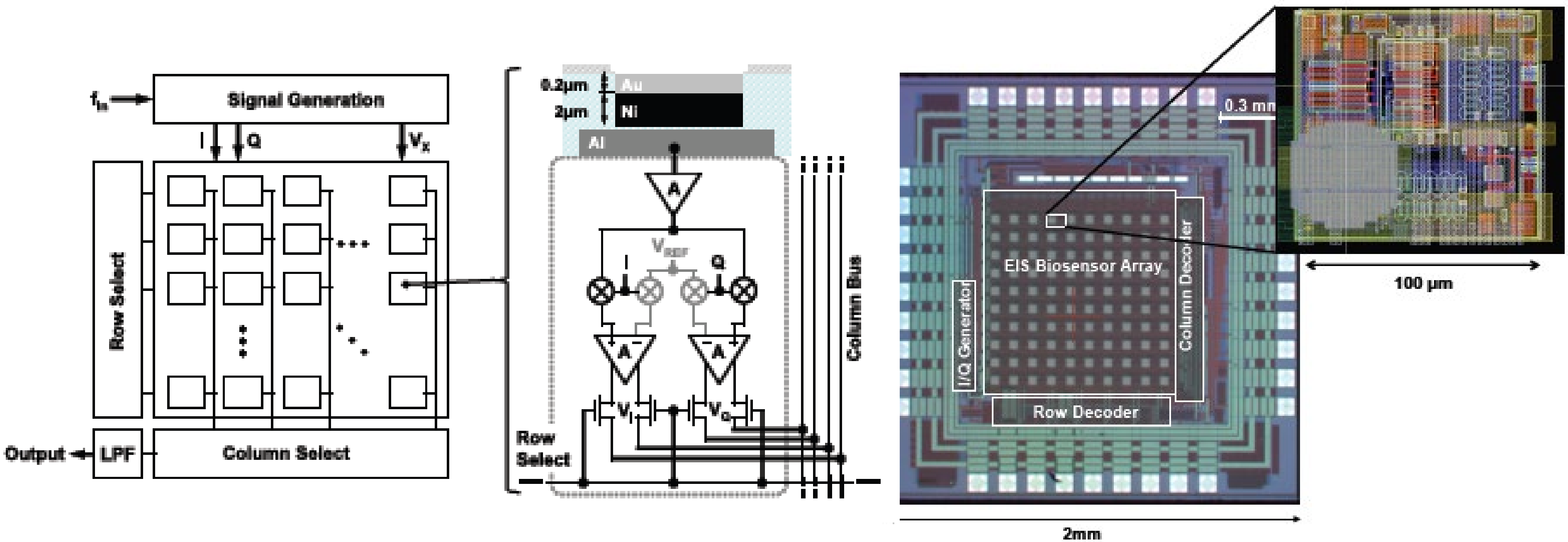




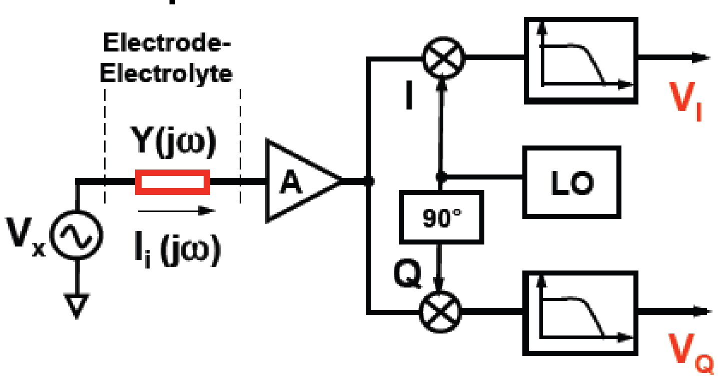
Polypyrrole

F. Heer *et al.*
ISSCC 2008

Ex 3: Impedande Affinity Biosensors (100 Ch.)

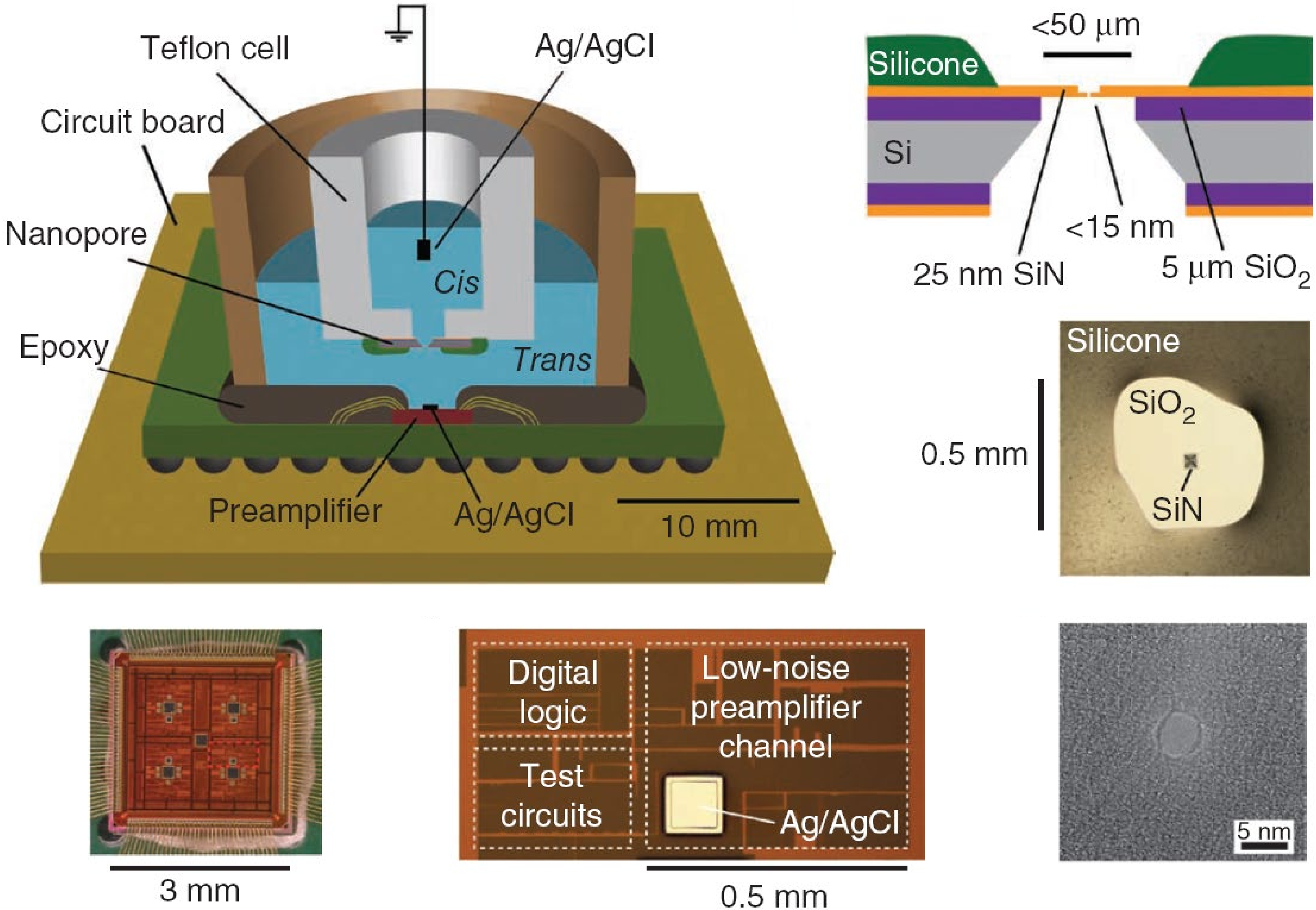


Impedance reader



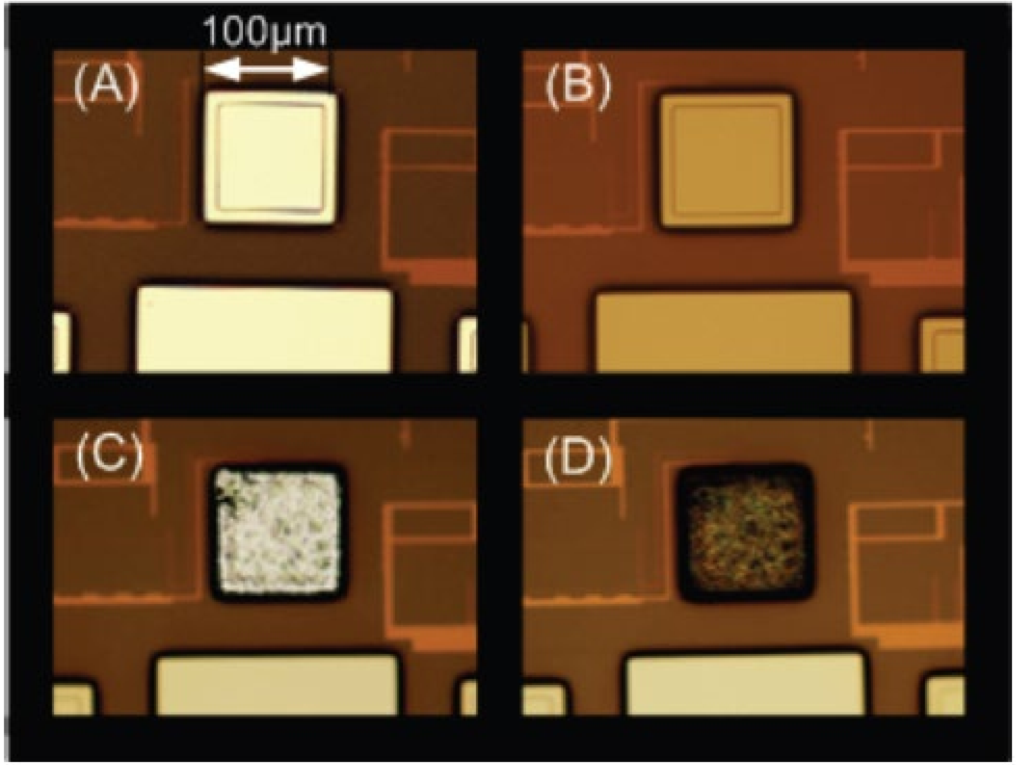
Technology	0.35 μm CMOS, 4 metal layers, 3.3 V supply
Die size	2 mm × 2 mm
Array size	10 × 10
Pixel size	100 μm × 100 μm
Electrode	40 μm × 40 μm, Al/1% Si
Frequency range	10 Hz–50 MHz
Power consumption	84.8 mW (100 kHz)
Dynamic Range (BW = 10 Hz)	97 dB

Ex 4: Nanopore Setup Optimization



Alchemy: Turning Aluminum in Silver

Chemical modification of the **pad** to serve as reference electrode:



Al

(A) as received from the fab
(B) chemical removal of Al
(C) Ag electroplating 10µm
(D) chemical chlorination

Ag/AgCl

Very useful post-CMOS electrochemical processing: no masks!

Although, mask used to open extra 300µm SU-8 insulation layer

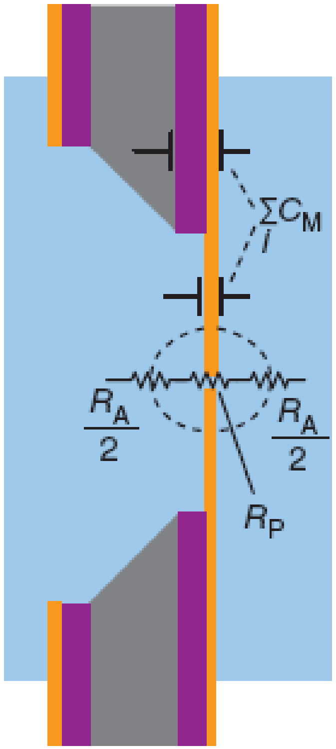
Careful Reduction of Parasitics

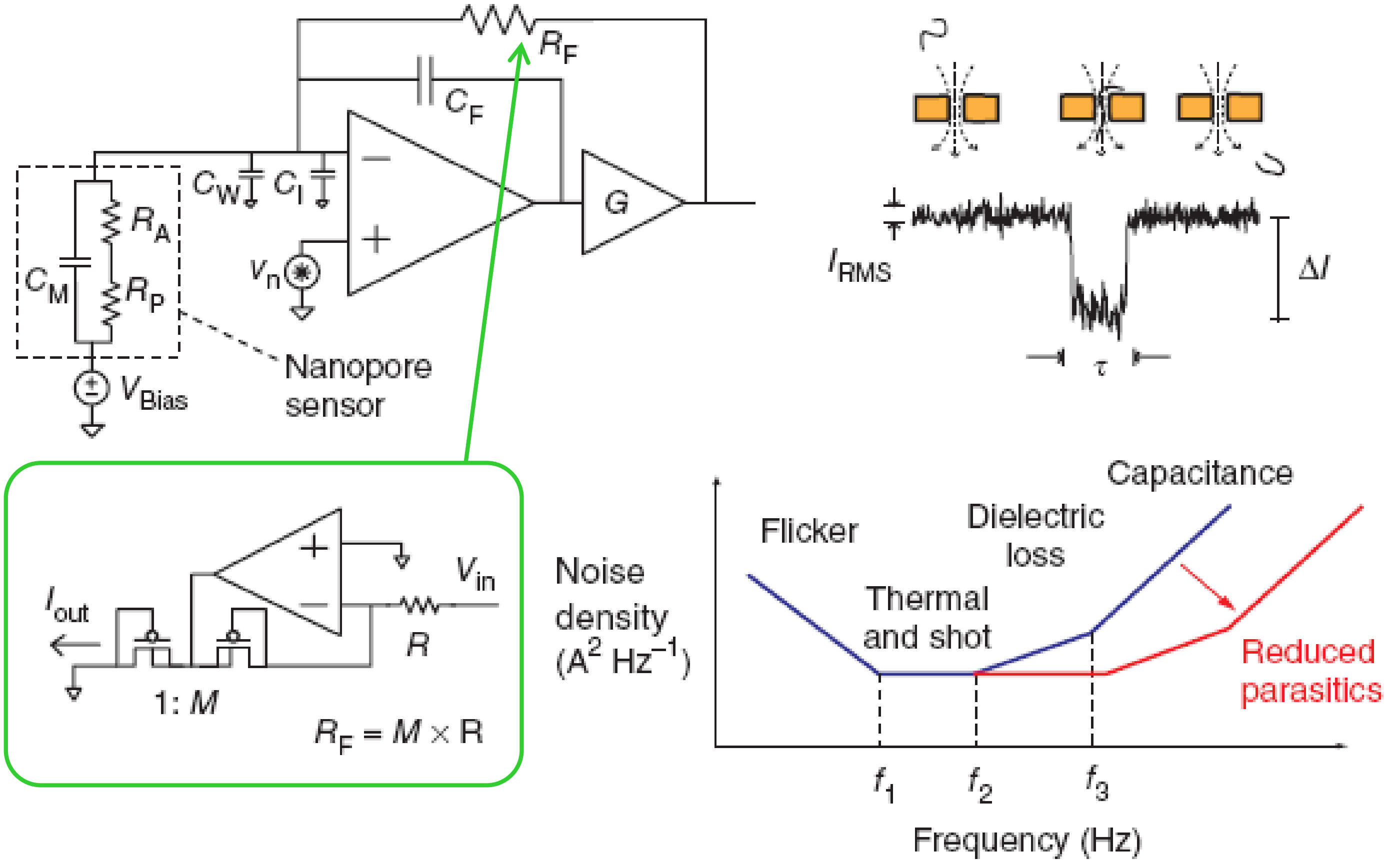
Device level:

Thick insulation
Small area

Circuit level:

State-of-the-art low-noise analog front-end





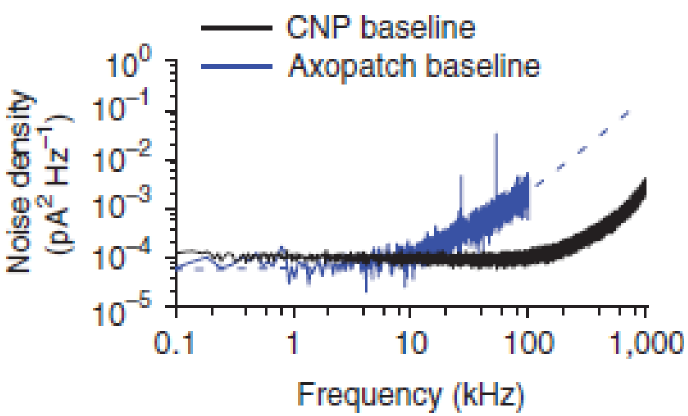
Achieved Significant Cap Reduction

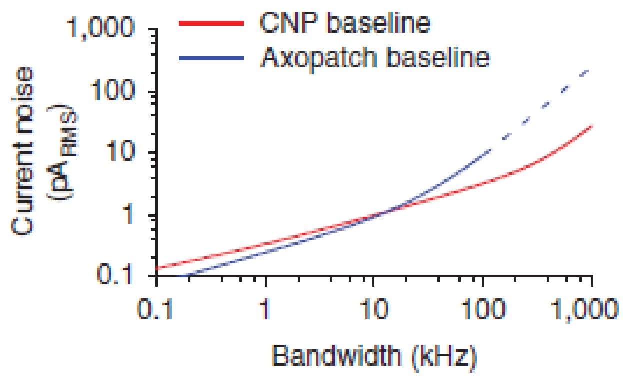
	C_I amplifier input	C_F amplifier feedback	C_W wiring/ interconnect/ fluidics	C_M solid-state nanopore chip/membrane	Total $C_I+C_F+C_W+C_M$
Axopatch 200B (early solid-state pores[3])	15 pF	1 pF	4 pF	300 pF	320 pF
Axopatch 200B (lower-capacitance solid-state pores[4,5])				10 pF	30 pF
This work (CNP, "PoreA")	1 pF	0.15 pF	0.25 pF	6 pF	7.4 pF

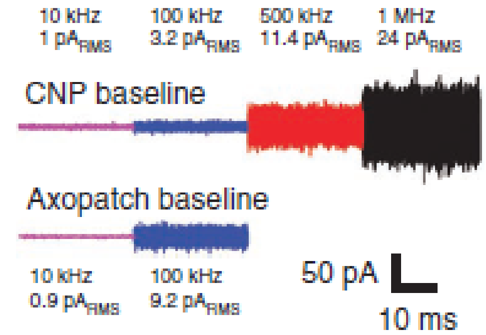
	Area	Thickness	Relative Dielectric Constant (ε_r)	Capacitance (C_M)
Ultra-thin SiN	$(500\ \text{nm})^2 = 2.5\text{x}10^{-7}\ \text{mm}^2$	10 nm	7	0.002 pF
SiN membrane	$(40\ \mu\text{m})^2 = 0.0016\ \text{mm}^2$	25 nm	7	4 pF
SiN-SiO_2 exposed to *trans* chamber	$\pi/4\ \text{x}\ (450\ \mu\text{m})^2 = 0.16\ \text{mm}^2$	5 µm	4	1.1 pF
Silicone-SiN-SiO_2	$(5\ \text{mm})^2 = 25\ \text{mm}^2$	1 mm	4	0.9 pF
			TOTAL	6 pF

Results: Characterization

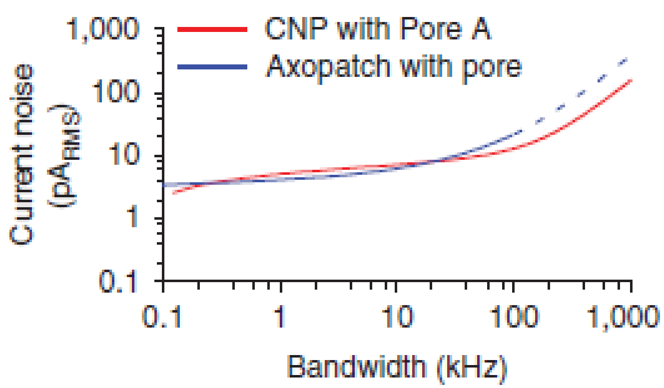
Without pore (min cap.):

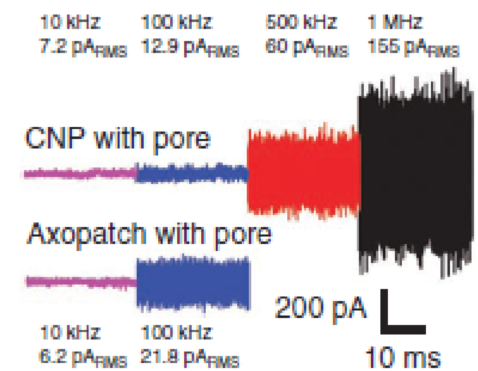






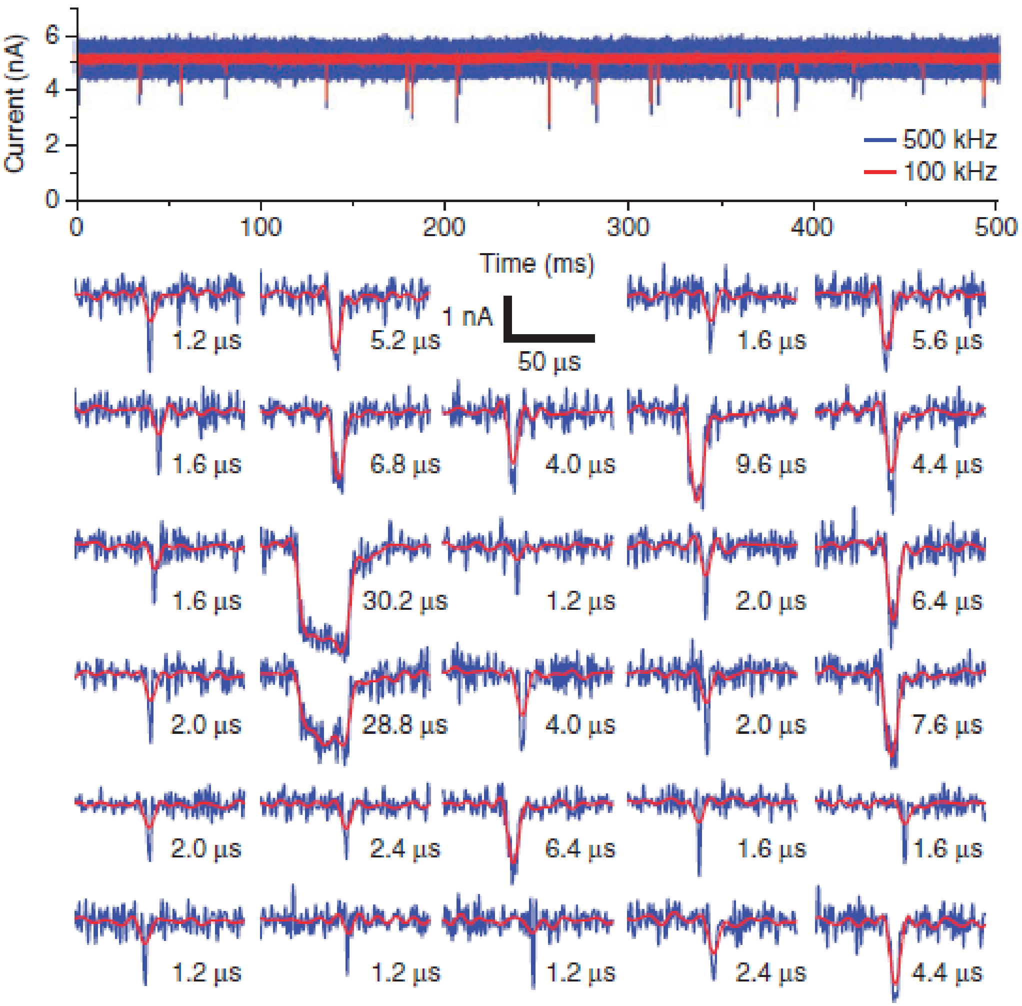
With pore connected:



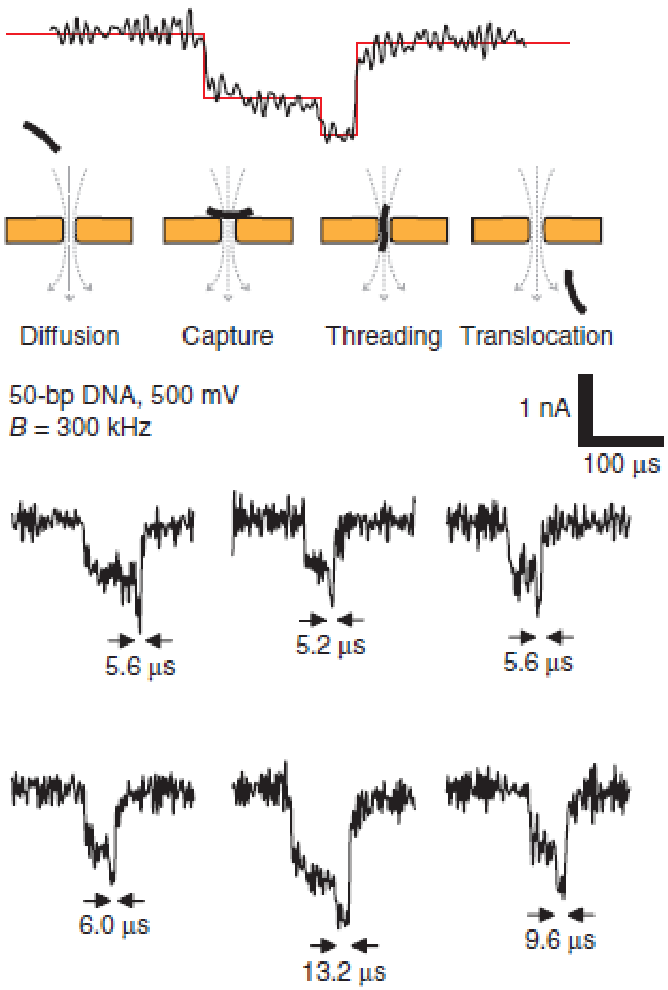


Results: Fast DNA Detection

High speed (short 25bp dsDNA)



High resolution
(intra-event dynamics)



References

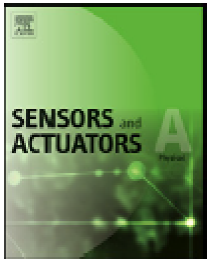
Sensors and Actuators A 172 (2011) 117–123

ELSEVIER

Contents lists available at ScienceDirect

Sensors and Actuators A: Physical

journal homepage: www.elsevier.com/locate/sna



ZeptoFarad capacitance detection with a miniaturized CMOS current front-end for nanoscale sensors

Marco Carminati*, Giorgio Ferrari, Filippo Guagliardo, Marco Sampietro

Integrated nanopore sensing platform with sub-microsecond temporal resolution

Jacob K Rosenstein[1], Meni Wanunu[2,3], Christopher A Merchant[3], Marija Drndic[3] & Kenneth L Shepard[1]